

Laboratory Work 2

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Chapter 1

Namespace Index

1.1 Namespace List

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2.1 Class Hierarchy

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Class Index

3.1 Class List

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Chapter 5

Namespace Documentation

5.1 Constants Namespace Reference

Variables

- constexpr int [min_attack_bonus](#) { 1 }
- constexpr int [max_attack_bonus](#) { 5 }
- constexpr int [min_crit_multiplier](#) { 200 }
- constexpr int [max_crit_multiplier](#) { 299 }
- constexpr int [critThreshold](#) { 3 }
- constexpr std::array [statThreshold](#) { 6, 8, 10, 13, 16, 20 }
- constexpr double [low_bonus](#) { -0.1 }
- constexpr double [high_bonus](#) { 0.2 }
- constexpr int [min_armor_reduce](#) { 40 }
- constexpr int [max_armor_reduce](#) { 60 }
- constexpr int [min_block_reduce](#) { 50 }
- constexpr int [max_block_reduce](#) { 100 }
- constexpr int [min_weapon_decoy](#) { 2 }
- constexpr int [max_weapon_decoy](#) { 5 }
- constexpr int [points_to_distribute](#) { 10 }
- constexpr int [initial_ap](#) { 10 }
- constexpr int [concentrationManaRestore](#) { 10 }
- constexpr int [healAmount](#) { 20 }
- constexpr size_t [maxInventorySize](#) { 5 }
- constexpr int [block_cost](#) { 5 }
- constexpr int [melee_cost](#) { 5 }
- constexpr int [cast_cost](#) { 3 }
- constexpr int [invest_cost](#) { 2 }
- constexpr int [conc_cost](#) { 5 }
- constexpr int [heal_cost](#) { 2 }

5.1.1 Variable Documentation

5.1.1.1 block_cost

```
constexpr int Constants::block_cost { 5 } [inline], [constexpr]
```

5.1.1.2 cast_cost

```
constexpr int Constants::cast_cost { 3 } [inline], [constexpr]
```

5.1.1.3 conc_cost

```
constexpr int Constants::conc_cost { 5 } [inline], [constexpr]
```

5.1.1.4 concentrationManaRestore

```
constexpr int Constants::concentrationManaRestore { 10 } [inline], [constexpr]
```

5.1.1.5 critThreshold

```
constexpr int Constants::critThreshold { 3 } [inline], [constexpr]
```

5.1.1.6 heal_cost

```
constexpr int Constants::heal_cost { 2 } [inline], [constexpr]
```

5.1.1.7 healAmount

```
constexpr int Constants::healAmount { 20 } [inline], [constexpr]
```

5.1.1.8 high_bonus

```
constexpr double Constants::high_bonus { 0.2 } [inline], [constexpr]
```


5.1.1.9 initial_ap

```
constexpr int Constants::initial_ap { 10 } [inline], [constexpr]
```

5.1.1.10 invest_cost

```
constexpr int Constants::invest_cost { 2 } [inline], [constexpr]
```

5.1.1.11 low_bonus

```
constexpr double Constants::low_bonus { -0.1 } [inline], [constexpr]
```

5.1.1.12 max_armor_reduce

```
constexpr int Constants::max_armor_reduce { 60 } [inline], [constexpr]
```

5.1.1.13 max_attack_bonus

```
constexpr int Constants::max_attack_bonus { 5 } [inline], [constexpr]
```

5.1.1.14 max_block_reduce

```
constexpr int Constants::max_block_reduce { 100 } [inline], [constexpr]
```

5.1.1.15 max_crit_multiplier

```
constexpr int Constants::max_crit_multiplier { 299 } [inline], [constexpr]
```

5.1.1.16 max_weapon_decoy

```
constexpr int Constants::max_weapon_decoy { 5 } [inline], [constexpr]
```

5.1.1.17 maxInventorySize

```
constexpr size_t Constants::maxInventorySize { 5 } [inline], [constexpr]
```

5.1.1.18 melee_cost

```
constexpr int Constants::melee_cost { 5 } [inline], [constexpr]
```

5.1.1.19 min_armor_reduce

```
constexpr int Constants::min_armor_reduce { 40 } [inline], [constexpr]
```

5.1.1.20 min_attack_bonus

```
constexpr int Constants::min_attack_bonus { 1 } [inline], [constexpr]
```

5.1.1.21 min_block_reduce

```
constexpr int Constants::min_block_reduce { 50 } [inline], [constexpr]
```

5.1.1.22 min_crit_multiplier

```
constexpr int Constants::min_crit_multiplier { 200 } [inline], [constexpr]
```

5.1.1.23 min_weapon_decoy

```
constexpr int Constants::min_weapon_decoy { 2 } [inline], [constexpr]
```

5.1.1.24 points_to_distribute

```
constexpr int Constants::points_to_distribute { 10 } [inline], [constexpr]
```

5.1.1.25 statThreshold

```
constexpr std::array Constants::statThreshold { 6, 8, 10, 13, 16, 20 } [inline], [constexpr]
```

5.2 Random Namespace Reference

Functions

- `std::mt19937` [generate](#) ()
- `int` [get](#) (int min, int max)
- `template<typename T >`
`T` [get](#) (T min, T max)
- `template<typename R , typename S , typename T >`
`R` [get](#) (S min, T max)

Variables

- `std::mt19937` [mt](#) { [generate](#)() }

5.2.1 Function Documentation

5.2.1.1 generate()

```
std::mt19937 Random::generate ( ) [inline]
```

5.2.1.2 get() [1/3]

```
int Random::get (
    int min,
    int max ) [inline]
```

5.2.1.3 get() [2/3]

```
template<typename R , typename S , typename T >
R Random::get (
    S min,
    T max )
```

5.2.1.4 `get()` [3/3]

```
template<typename T >
T Random::get (
    T min,
    T max )
```

5.2.2 Variable Documentation

5.2.2.1 `mt`

```
std::mt19937 Random::mt { generate\(\) } [inline]
```

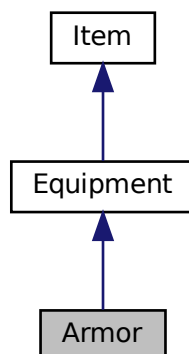
Chapter 6

Class Documentation

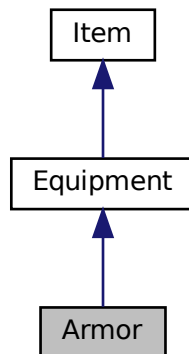
6.1 Armor Class Reference

```
#include <Armor.h>
```

Inheritance diagram for Armor:



Collaboration diagram for Armor:



Public Member Functions

- [Armor](#) (std::string_view name, int defense, int durability)
- void [use](#) ([Character](#) &user, std::unique_ptr< [Item](#) > [self](#)) override
- int [getEffectiveDefense](#) () const
- void [takeDamage](#) (int amount) override

Private Attributes

- int [baseDefense](#)

Additional Inherited Members

6.1.1 Constructor & Destructor Documentation

6.1.1.1 Armor()

```
Armor::Armor (
    std::string_view name,
    int defense,
    int durability )
```

6.1.2 Member Function Documentation

6.1.2.1 `getEffectiveDefense()`

```
int Armor::getEffectiveDefense ( ) const
```

6.1.2.2 `takeDamage()`

```
void Armor::takeDamage (
    int amount ) [override], [virtual]
```

Reimplemented from [Equipment](#).

6.1.2.3 `use()`

```
void Armor::use (
    Character & user,
    std::unique_ptr< Item > self ) [override], [virtual]
```

Implements [Equipment](#).

6.1.3 Member Data Documentation

6.1.3.1 `baseDefense`

```
int Armor::baseDefense [private]
```

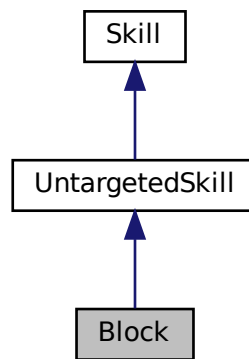
The documentation for this class was generated from the following files:

- `src/Equipment/Armor.h`
- `src/Equipment/Armor.cpp`

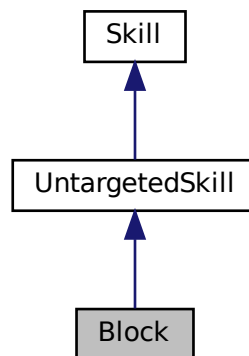
6.2 Block Class Reference

```
#include <Block.h>
```

Inheritance diagram for Block:



Collaboration diagram for Block:



Public Member Functions

- [Block](#) ()
- void [execute](#) ([Character](#) &user) override
- std::string_view [getName](#) () const override

Additional Inherited Members

6.2.1 Constructor & Destructor Documentation

6.2.1.1 Block()

```
Block::Block ( ) [inline]
```

6.2.2 Member Function Documentation

6.2.2.1 execute()

```
void Block::execute (
    Character & user ) [override], [virtual]
```

Implements [UntargetedSkill](#).

6.2.2.2 getName()

```
std::string_view Block::getName ( ) const [inline], [override], [virtual]
```

Implements [Skill](#).

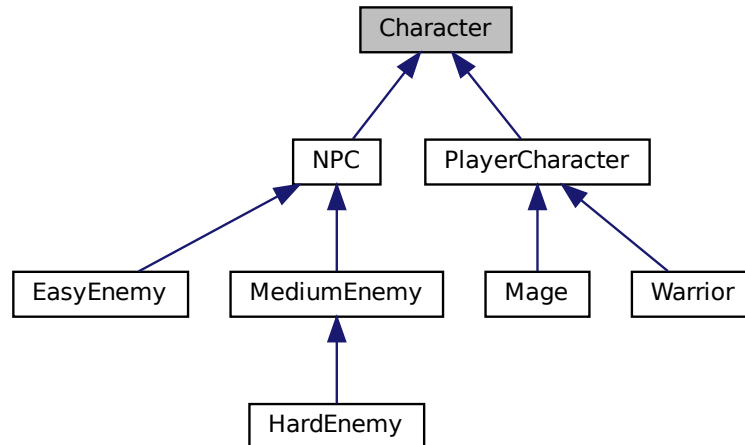
The documentation for this class was generated from the following files:

- [src/Skills/Block.h](#)
- [src/Skills/Block.cpp](#)

6.3 Character Class Reference

```
#include <Character.h>
```

Inheritance diagram for Character:



Public Member Functions

- `Character` (std::string_view name, int strength, int intelligence, int agility, int mana, int health=100, int actionPoints=10)
- `Character` (std::string_view name, int agility, int mana)
- `Character` ()=default
- virtual `~Character` ()=default
- void `setStats` (int strength, int intelligence)
- void `setBlocking` (bool state)
- bool `isBlocking` () const
- virtual void `resetActionPoints` ()
- void `incrementAttackCount` ()
- void `resetAttackCount` ()
- int `getAttackCount` () const
- const std::string & `getName` () const
- int `getStrength` () const
- int `getIntelligence` () const
- int `getHealth` () const
- int `getMaxHealth` () const
- int `getAgility` () const
- int `getActionPoints` () const
- int `getMana` () const
- bool `isDead` () const
- void `increaseHealth` (int amount)
- void `increaseMana` (int amount)
- template<typename T >
void `modifyStat` (T Character::*statPtr, T delta, const std::string &statName)

- void `increaseMaxHealth` (int amount)
- void `increaseMaxMana` (int amount)
- void `increaseStrength` (int amount)
- void `increaseIntelligence` (int amount)
- void `increaseAgility` (int amount)
- void `increaseActionPoints` (int amount)
- void `repairArmor` (int durability)
- void `repairWeapon` (int durability)
- virtual void `displayStatus` () const =0
- void `equipWeapon` (std::unique_ptr< `Weapon` > `weapon`)
- void `equipArmor` (std::unique_ptr< `Armor` > `armor`)
- void `takeDamage` (`DamageCalculator` &dmgCalc)
- void `takeDamage` (int `damage`)
- void `takeActionPoints` (int cost)
- void `useMana` (int cost)
- `Weapon` * `getWeapon` () const
- `Armor` * `getArmor` () const

Protected Attributes

- std::string `m_name`
- int `m_strength` { 5 }
- int `m_intelligence` { 5 }
- int `m_agility` { 5 }
- int `m_mana` { 50 }
- int `m_maxMana` { 50 }
- int `m_health` { 100 }
- int `m_maxHealth` { 100 }
- int `m_actionPoints` { `Constants::initial_ap` }
- int `m_maxActionPoints` { `Constants::initial_ap` }
- bool `m_isBlocking` { false }
- int `m_attackCount` { 0 }
- std::unique_ptr< `Weapon` > `equippedWeapon`
- std::unique_ptr< `Armor` > `equippedArmor`

6.3.1 Constructor & Destructor Documentation

6.3.1.1 `Character()` [1/3]

```
Character::Character (
    std::string_view name,
    int strength,
    int intelligence,
    int agility,
    int mana,
    int health = 100,
    int actionPoints = 10 )
```

6.3.1.2 Character() [2/3]

```
Character::Character (
    std::string_view name,
    int agility,
    int mana )
```

6.3.1.3 Character() [3/3]

```
Character::Character ( ) [default]
```

6.3.1.4 ~Character()

```
virtual Character::~~Character ( ) [virtual], [default]
```

6.3.2 Member Function Documentation

6.3.2.1 displayStatus()

```
virtual void Character::displayStatus ( ) const [pure virtual]
```

Implemented in [PlayerCharacter](#), and [NPC](#).

6.3.2.2 equipArmor()

```
void Character::equipArmor (
    std::unique_ptr< Armor > armor )
```

6.3.2.3 equipWeapon()

```
void Character::equipWeapon (
    std::unique_ptr< Weapon > weapon )
```

6.3.2.4 `getActionPoints()`

```
int Character::getActionPoints ( ) const
```

6.3.2.5 `getAgility()`

```
int Character::getAgility ( ) const
```

6.3.2.6 `getArmor()`

```
Armor* Character::getArmor ( ) const [inline]
```

6.3.2.7 `getAttackCount()`

```
int Character::getAttackCount ( ) const
```

6.3.2.8 `getHealth()`

```
int Character::getHealth ( ) const
```

6.3.2.9 `getIntelligence()`

```
int Character::getIntelligence ( ) const
```

6.3.2.10 `getMana()`

```
int Character::getMana ( ) const
```

6.3.2.11 `getMaxHealth()`

```
int Character::getMaxHealth ( ) const
```

6.3.2.12 getName()

```
const std::string & Character::getName ( ) const
```

6.3.2.13 getStrength()

```
int Character::getStrength ( ) const
```

6.3.2.14 getWeapon()

```
Weapon* Character::getWeapon ( ) const [inline]
```

6.3.2.15 increaseActionPoints()

```
void Character::increaseActionPoints (
    int amount ) [inline]
```

6.3.2.16 increaseAgility()

```
void Character::increaseAgility (
    int amount ) [inline]
```

6.3.2.17 increaseHealth()

```
void Character::increaseHealth (
    int amount )
```

6.3.2.18 increaseIntelligence()

```
void Character::increaseIntelligence (
    int amount ) [inline]
```

6.3.2.19 increaseMana()

```
void Character::increaseMana (
    int amount )
```

6.3.2.20 increaseMaxHealth()

```
void Character::increaseMaxHealth (
    int amount ) [inline]
```

6.3.2.21 increaseMaxMana()

```
void Character::increaseMaxMana (
    int amount ) [inline]
```

6.3.2.22 increaseStrength()

```
void Character::increaseStrength (
    int amount ) [inline]
```

6.3.2.23 incrementAttackCount()

```
void Character::incrementAttackCount ( )
```

6.3.2.24 isBlocking()

```
bool Character::isBlocking ( ) const
```

6.3.2.25 isDead()

```
bool Character::isDead ( ) const
```

6.3.2.26 modifyStat()

```
template<typename T >
void Character::modifyStat (
    T Character::* statPtr,
    T delta,
    const std::string & statName ) [inline]
```

6.3.2.27 repairArmor()

```
void Character::repairArmor (
    int durability )
```

6.3.2.28 repairWeapon()

```
void Character::repairWeapon (
    int durability )
```

6.3.2.29 resetActionPoints()

```
void Character::resetActionPoints ( ) [virtual]
```

Reimplemented in [Warrior](#).

6.3.2.30 resetAttackCount()

```
void Character::resetAttackCount ( )
```

6.3.2.31 setBlocking()

```
void Character::setBlocking (
    bool state )
```


6.3.2.32 setStats()

```
void Character::setStats (
    int strength,
    int intelligence )
```

6.3.2.33 takeActionPoints()

```
void Character::takeActionPoints (
    int cost )
```

6.3.2.34 takeDamage() [1/2]

```
void Character::takeDamage (
    DamageCalculator & dmgCalc )
```

6.3.2.35 takeDamage() [2/2]

```
void Character::takeDamage (
    int damage )
```

6.3.2.36 useMana()

```
void Character::useMana (
    int cost )
```

6.3.3 Member Data Documentation

6.3.3.1 equippedArmor

```
std::unique_ptr<Armor> Character::equippedArmor [protected]
```

6.3.3.2 equippedWeapon

```
std::unique_ptr<Weapon> Character::equippedWeapon [protected]
```

6.3.3.3 m_actionPoints

```
int Character::m_actionPoints { Constants::initial_ap } [protected]
```

6.3.3.4 m_agility

```
int Character::m_agility { 5 } [protected]
```

6.3.3.5 m_attackCount

```
int Character::m_attackCount { 0 } [protected]
```

6.3.3.6 m_health

```
int Character::m_health { 100 } [protected]
```

6.3.3.7 m_intelligence

```
int Character::m_intelligence { 5 } [protected]
```

6.3.3.8 m_isBlocking

```
bool Character::m_isBlocking { false } [protected]
```

6.3.3.9 m_mana

```
int Character::m_mana { 50 } [protected]
```

6.3.3.10 m_maxActionPoints

```
int Character::m_maxActionPoints { Constants::initial_ap } [protected]
```

6.3.3.11 m_maxHealth

```
int Character::m_maxHealth { 100 } [protected]
```

6.3.3.12 m_maxMana

```
int Character::m_maxMana { 50 } [protected]
```

6.3.3.13 m_name

```
std::string Character::m_name [protected]
```

6.3.3.14 m_strength

```
int Character::m_strength { 5 } [protected]
```

The documentation for this class was generated from the following files:

- src/Character/[Character.h](#)
- src/Character/[Character.cpp](#)

6.4 CharacterCreation Class Reference

```
#include <CharacterCreation.h>
```

Public Member Functions

- `std::unique_ptr< PlayerCharacter > createCharacter ()`

Static Protected Member Functions

- static std::string [ChooseName](#) ()
- static std::unique_ptr< [PlayerCharacter](#) > [ChooseClass](#) (std::string_view name)
- static void [CustomizeStats](#) ([PlayerCharacter](#) &character)
- static void [ChooseWeapon](#) ([PlayerCharacter](#) &character)
- static void [ChooseArmor](#) ([PlayerCharacter](#) &character)
- static void [ChooseSkills](#) ([PlayerCharacter](#) &character)
- static void [ChooseInventory](#) ([PlayerCharacter](#) &character)

6.4.1 Member Function Documentation

6.4.1.1 ChooseArmor()

```
void CharacterCreation::ChooseArmor (  
    PlayerCharacter & character ) [static], [protected]
```

6.4.1.2 ChooseClass()

```
std::unique_ptr< PlayerCharacter > CharacterCreation::ChooseClass (  
    std::string_view name ) [static], [protected]
```

6.4.1.3 ChooseInventory()

```
void CharacterCreation::ChooseInventory (  
    PlayerCharacter & character ) [static], [protected]
```

6.4.1.4 ChooseName()

```
std::string CharacterCreation::ChooseName ( ) [static], [protected]
```

6.4.1.5 ChooseSkills()

```
void CharacterCreation::ChooseSkills (  
    PlayerCharacter & character ) [static], [protected]
```

6.4.1.6 ChooseWeapon()

```
void CharacterCreation::ChooseWeapon (
    PlayerCharacter & character ) [static], [protected]
```

6.4.1.7 createCharacter()

```
std::unique_ptr< PlayerCharacter > CharacterCreation::createCharacter ( )
```

6.4.1.8 CustomizeStats()

```
void CharacterCreation::CustomizeStats (
    PlayerCharacter & character ) [static], [protected]
```

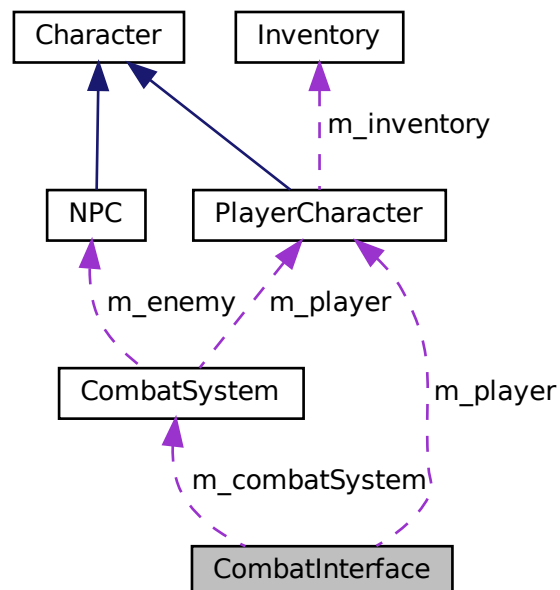
The documentation for this class was generated from the following files:

- [src/Game/CharacterCreation.h](#)
- [src/Game/CharacterCreation.cpp](#)

6.5 CombatInterface Class Reference

```
#include <CombatInterface.h>
```

Collaboration diagram for CombatInterface:



Public Member Functions

- [CombatInterface](#) ([PlayerCharacter](#) &player, [CombatSystem](#) &combatSystem)
- void [displayCombatOptions](#) ()
- void [processInput](#) (int choice)
- void [performAttack](#) ()
- void [openInventory](#) ()

Private Attributes

- [PlayerCharacter](#) & m_player
- [CombatSystem](#) & m_combatSystem

6.5.1 Constructor & Destructor Documentation

6.5.1.1 CombatInterface()

```
CombatInterface::CombatInterface (  
    PlayerCharacter & player,  
    CombatSystem & combatSystem )
```

6.5.2 Member Function Documentation

6.5.2.1 displayCombatOptions()

```
void CombatInterface::displayCombatOptions ( )
```

6.5.2.2 openInventory()

```
void CombatInterface::openInventory ( )
```

6.5.2.3 performAttack()

```
void CombatInterface::performAttack ( )
```

6.5.2.4 processInput()

```
void CombatInterface::processInput (
    int choice )
```

6.5.3 Member Data Documentation

6.5.3.1 m_combatSystem

```
CombatSystem& CombatInterface::m_combatSystem [private]
```

6.5.3.2 m_player

```
PlayerCharacter& CombatInterface::m_player [private]
```

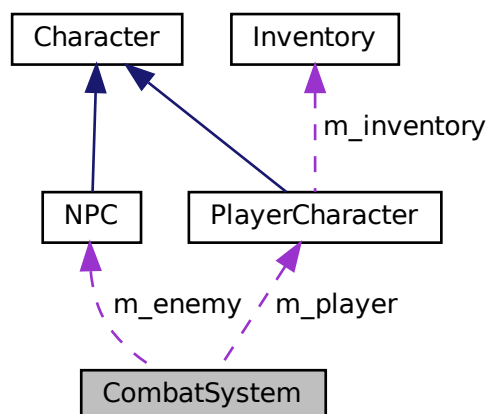
The documentation for this class was generated from the following files:

- src/Game/CombatInterface.h
- src/Game/CombatInterface.cpp

6.6 CombatSystem Class Reference

```
#include <CombatSystem.h>
```

Collaboration diagram for CombatSystem:



Public Member Functions

- [CombatSystem](#) ([PlayerCharacter](#) &player, [NPC](#) &enemy)
- void [startCombat](#) ([CombatInterface](#) &interface)
- void [handleTurn](#) ([CombatInterface](#) &interface)
- [Character](#) & [getEnemy](#) ()
- bool [isCombatOver](#) () const
- bool [playerLost](#) () const

Private Attributes

- [PlayerCharacter](#) & [m_player](#)
- [NPC](#) * [m_enemy](#)
- bool [m_playerTurn](#) {}

6.6.1 Constructor & Destructor Documentation

6.6.1.1 CombatSystem()

```
CombatSystem::CombatSystem (
    PlayerCharacter & player,
    NPC & enemy )
```

6.6.2 Member Function Documentation

6.6.2.1 getEnemy()

```
Character& CombatSystem::getEnemy ( ) [inline]
```

6.6.2.2 handleTurn()

```
void CombatSystem::handleTurn (
    CombatInterface & interface )
```

6.6.2.3 isCombatOver()

```
bool CombatSystem::isCombatOver ( ) const
```


6.6.2.4 playerLost()

```
bool CombatSystem::playerLost ( ) const
```

6.6.2.5 startCombat()

```
void CombatSystem::startCombat (
    CombatInterface & interface )
```

6.6.3 Member Data Documentation

6.6.3.1 m_enemy

```
NPC\* CombatSystem::m_enemy [private]
```

6.6.3.2 m_player

```
PlayerCharacter& CombatSystem::m_player [private]
```

6.6.3.3 m_playerTurn

```
bool CombatSystem::m_playerTurn {} [private]
```

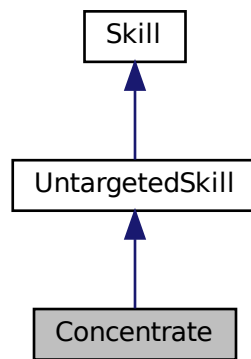
The documentation for this class was generated from the following files:

- [src/Game/CombatSystem.h](#)
- [src/Game/CombatSystem.cpp](#)

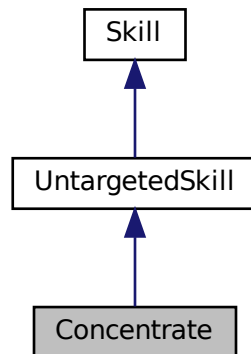
6.7 Concentrate Class Reference

```
#include <Concentrate.h>
```

Inheritance diagram for Concentrate:



Collaboration diagram for Concentrate:



Public Member Functions

- [Concentrate](#) ()
- void [execute](#) ([Character](#) &user) override
- std::string_view [getName](#) () const override

Additional Inherited Members

6.7.1 Constructor & Destructor Documentation

6.7.1.1 Concentrate()

```
Concentrate::Concentrate ( ) [inline]
```

6.7.2 Member Function Documentation

6.7.2.1 execute()

```
void Concentrate::execute (
    Character & user ) [override], [virtual]
```

Implements [UntargetedSkill](#).

6.7.2.2 getName()

```
std::string_view Concentrate::getName ( ) const [inline], [override], [virtual]
```

Implements [Skill](#).

The documentation for this class was generated from the following files:

- [src/Skills/Concentrate.h](#)
- [src/Skills/Concentrate.cpp](#)

6.8 DamageCalculator Class Reference

```
#include <DamageCalculator.h>
```

Public Member Functions

- [DamageCalculator](#) ()=default
- void [markAsSpell](#) ()
- bool [isSpell](#) () const

Private Attributes

- int `m_finalDamage` { 0 }
- int `m_armorDamage` { 0 }
- bool `m_isSpell` { false }

Friends

- void `calculateDamage` (`DamageCalculator` &dmgCalc, `Character` &target, int baseDamage, int statBonus, int attackCount)
- void `Character::takeDamage` (`DamageCalculator` &dmgCalc)

6.8.1 Constructor & Destructor Documentation

6.8.1.1 DamageCalculator()

```
DamageCalculator::DamageCalculator ( ) [default]
```

6.8.2 Member Function Documentation

6.8.2.1 isSpell()

```
bool DamageCalculator::isSpell ( ) const [inline]
```

6.8.2.2 markAsSpell()

```
void DamageCalculator::markAsSpell ( ) [inline]
```

6.8.3 Friends And Related Function Documentation

6.8.3.1 calculateDamage

```
void calculateDamage (
    DamageCalculator & dmgCalc,
    Character & target,
    int baseDamage,
    int statBonus,
    int attackCount ) [friend]
```

6.8.3.2 Character::takeDamage

```
void Character::takeDamage (
    DamageCalculator & dmgCalc ) [friend]
```

6.8.4 Member Data Documentation

6.8.4.1 m_armorDamage

```
int DamageCalculator::m_armorDamage { 0 } [private]
```

6.8.4.2 m_finalDamage

```
int DamageCalculator::m_finalDamage { 0 } [private]
```

6.8.4.3 m_isSpell

```
bool DamageCalculator::m_isSpell { false } [private]
```

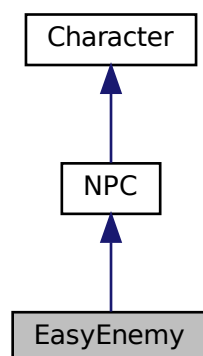
The documentation for this class was generated from the following file:

- [src/Game/DamageCalculator.h](#)

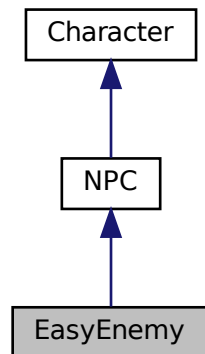
6.9 EasyEnemy Class Reference

```
#include <EasyEnemy.h>
```

Inheritance diagram for EasyEnemy:



Collaboration diagram for EasyEnemy:



Public Member Functions

- [EasyEnemy](#) ()
- void [performTurn](#) ([PlayerCharacter](#) &player) override

Private Member Functions

- void [equipBackup](#) () override

Private Attributes

- bool [m_usedUltra](#) { false }

Additional Inherited Members

6.9.1 Constructor & Destructor Documentation

6.9.1.1 EasyEnemy()

```
EasyEnemy::EasyEnemy ( )
```

6.9.2 Member Function Documentation

6.9.2.1 equipBackup()

```
void EasyEnemy::equipBackup ( ) [override], [private], [virtual]
```

Implements [NPC](#).

6.9.2.2 performTurn()

```
void EasyEnemy::performTurn (
    PlayerCharacter & player ) [override], [virtual]
```

Implements [NPC](#).

6.9.3 Member Data Documentation

6.9.3.1 m_usedUltra

```
bool EasyEnemy::m_usedUltra { false } [private]
```

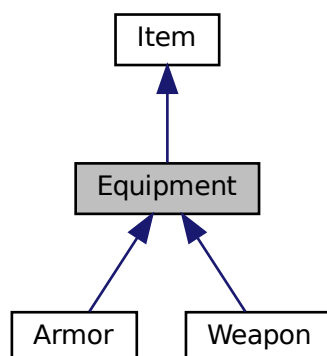
The documentation for this class was generated from the following files:

- [src/Character/EasyEnemy.h](#)
- [src/Character/EasyEnemy.cpp](#)

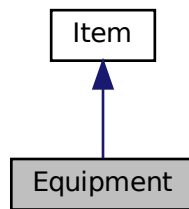
6.10 Equipment Class Reference

```
#include <Equipment.h>
```

Inheritance diagram for Equipment:



Collaboration diagram for Equipment:



Public Member Functions

- `Equipment` (`std::string_view` name, `Rarity` rarity, `int` maxDurability)
- virtual void `use` (`Character` &user, `std::unique_ptr`< `Item` > self)=0
- virtual void `takeDamage` (`int` amount)
- void `repair` (`int` amount)
- bool `isBroken` () const
- int `getCurrentDurability` () const
- int `getMaxDurability` () const
- double `getCondition` () const

Protected Attributes

- int `m_maxDurability`
- int `m_currentDurability`

6.10.1 Constructor & Destructor Documentation

6.10.1.1 Equipment()

```
Equipment::Equipment (
    std::string_view name,
    Rarity rarity,
    int maxDurability )
```

6.10.2 Member Function Documentation

6.10.2.1 `getCondition()`

```
double Equipment::getCondition ( ) const
```

6.10.2.2 `getCurrentDurability()`

```
int Equipment::getCurrentDurability ( ) const
```

6.10.2.3 `getMaxDurability()`

```
int Equipment::getMaxDurability ( ) const
```

6.10.2.4 `isBroken()`

```
bool Equipment::isBroken ( ) const
```

6.10.2.5 `repair()`

```
void Equipment::repair (
    int amount )
```

6.10.2.6 `takeDamage()`

```
void Equipment::takeDamage (
    int amount ) [virtual]
```

Reimplemented in [Armor](#).

6.10.2.7 `use()`

```
virtual void Equipment::use (
    Character & user,
    std::unique_ptr< Item > self ) [pure virtual]
```

Implemented in [Weapon](#), and [Armor](#).

6.10.3 Member Data Documentation

6.10.3.1 m_currentDurability

```
int Equipment::m_currentDurability [protected]
```

6.10.3.2 m_maxDurability

```
int Equipment::m_maxDurability [protected]
```

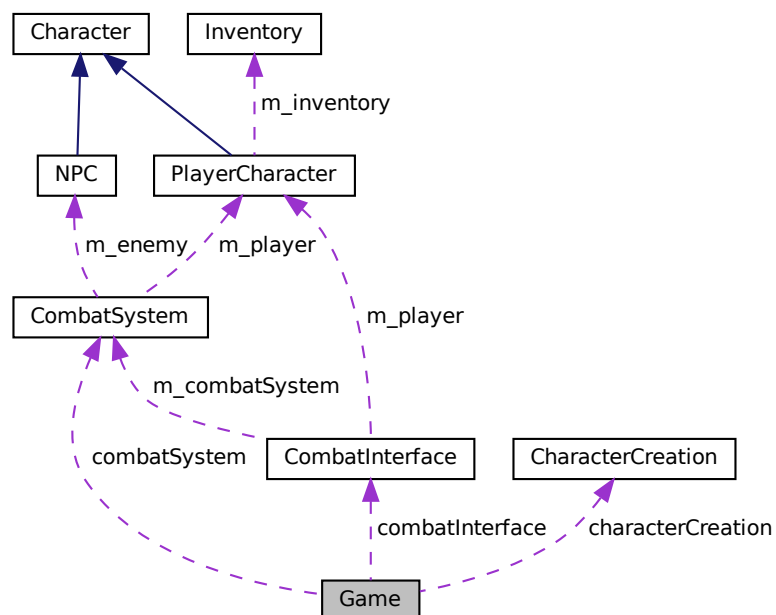
The documentation for this class was generated from the following files:

- [src/Equipment/Equipment.h](#)
- [src/Equipment/Equipment.cpp](#)

6.11 Game Class Reference

```
#include <Game.h>
```

Collaboration diagram for Game:



Public Member Functions

- [Game](#) ()=default
- [~Game](#) ()
- void [start](#) ()
- void [startCombat](#) ([NPC](#) &[enemy](#))

Private Attributes

- std::unique_ptr< [PlayerCharacter](#) > [player](#) {}
- std::unique_ptr< [NPC](#) > [enemy](#) {}
- [CharacterCreation](#) [characterCreation](#)
- [CombatSystem](#) * [combatSystem](#) {}
- [CombatInterface](#) * [combatInterface](#) {}

6.11.1 Constructor & Destructor Documentation

6.11.1.1 Game()

```
Game::Game ( ) [default]
```

6.11.1.2 ~Game()

```
Game::~~Game ( )
```

6.11.2 Member Function Documentation

6.11.2.1 start()

```
void Game::start ( )
```

6.11.2.2 startCombat()

```
void Game::startCombat (
    NPC & enemy )
```

6.11.3 Member Data Documentation

6.11.3.1 characterCreation

```
CharacterCreation Game::characterCreation [private]
```

6.11.3.2 combatInterface

```
CombatInterface* Game::combatInterface {} [private]
```

6.11.3.3 combatSystem

```
CombatSystem* Game::combatSystem {} [private]
```

6.11.3.4 enemy

```
std::unique_ptr<NPC> Game::enemy {} [private]
```

6.11.3.5 player

```
std::unique_ptr<PlayerCharacter> Game::player {} [private]
```

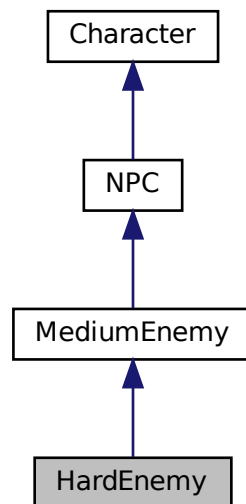
The documentation for this class was generated from the following files:

- [src/Game/Game.h](#)
- [src/Game/Game.cpp](#)

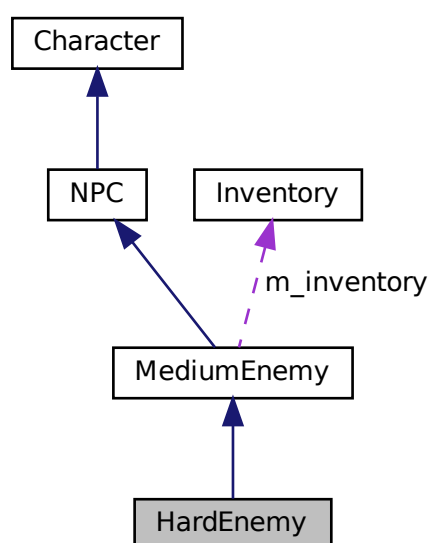
6.12 HardEnemy Class Reference

```
#include <HardEnemy.h>
```

Inheritance diagram for HardEnemy:



Collaboration diagram for HardEnemy:



Public Member Functions

- [HardEnemy](#) ()
- void [performTurn](#) ([PlayerCharacter](#) &player) override

Private Member Functions

- void [equipBackup](#) () override
- bool [shouldBlock](#) (const [PlayerCharacter](#) &player, double enemyHealthRatio)
- void [performBlock](#) ()

Additional Inherited Members

6.12.1 Constructor & Destructor Documentation

6.12.1.1 [HardEnemy\(\)](#)

```
HardEnemy::HardEnemy ( )
```

6.12.2 Member Function Documentation

6.12.2.1 [equipBackup\(\)](#)

```
void HardEnemy::equipBackup ( ) [override], [private], [virtual]
```

Implements [NPC](#).

6.12.2.2 [performBlock\(\)](#)

```
void HardEnemy::performBlock ( ) [private]
```

6.12.2.3 [performTurn\(\)](#)

```
void HardEnemy::performTurn (
    PlayerCharacter & player ) [override], [virtual]
```

Implements [NPC](#).

6.12.2.4 shouldBlock()

```
bool HardEnemy::shouldBlock (
    const PlayerCharacter & player,
    double enemyHealthRatio ) [private]
```

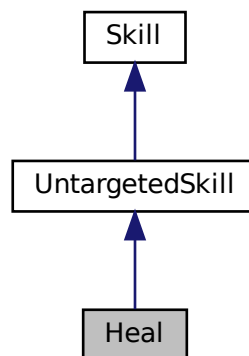
The documentation for this class was generated from the following files:

- [src/Character/HardEnemy.h](#)
- [src/Character/HardEnemy.cpp](#)

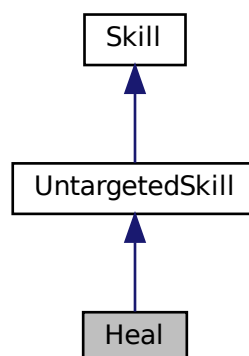
6.13 Heal Class Reference

```
#include <Heal.h>
```

Inheritance diagram for Heal:



Collaboration diagram for Heal:



Public Member Functions

- [Heal](#) (int cost=[Constants::heal_cost](#), int healAmount=[Constants::healAmount](#))
- void [execute](#) ([Character](#) &user) override
- int [getHealAmount](#) () const
- std::string_view [getName](#) () const override

Private Attributes

- int [m_healAmount](#) {}

Additional Inherited Members

6.13.1 Constructor & Destructor Documentation

6.13.1.1 Heal()

```
Heal::Heal (
    int cost = Constants::heal\_cost,
    int healAmount = Constants::healAmount )
```

6.13.2 Member Function Documentation

6.13.2.1 execute()

```
void Heal::execute (
    Character & user ) [override], [virtual]
```

Implements [UntargetedSkill](#).

6.13.2.2 getHealAmount()

```
int Heal::getHealAmount ( ) const [inline]
```

6.13.2.3 getName()

```
std::string_view Heal::getName ( ) const [inline], [override], [virtual]
```

Implements [Skill](#).

6.13.3 Member Data Documentation

6.13.3.1 m_healAmount

```
int Heal::m_healAmount {} [private]
```

The documentation for this class was generated from the following files:

- src/Skills/Heal.h
- src/Skills/Heal.cpp

6.14 Inventory Class Reference

```
#include <Inventory.h>
```

Public Member Functions

- int [getItemCount](#) () const
- bool [isFull](#) () const
- void [newSize](#) ()
- int [getSize](#) () const
- const [Item](#) * [operator\[\]](#) (int index) const
- void [addItem](#) (std::unique_ptr< [Item](#) > item)
- void [removeItem](#) (int index)
- void [removeBrokenItems](#) ()
- void [useItem](#) (int index, [Character](#) &user, [Character](#) *enemy=nullptr)
- void [display](#) () const
- bool [isEmpty](#) () const

Private Attributes

- std::vector< std::unique_ptr< [Item](#) > > [m_items](#)
- int [inventorySize](#) { 5 }

6.14.1 Member Function Documentation

6.14.1.1 addItem()

```
void Inventory::addItem (  
    std::unique_ptr< Item > item )
```

6.14.1.2 display()

```
void Inventory::display ( ) const
```

6.14.1.3 getItemCount()

```
int Inventory::getItemCount ( ) const
```

6.14.1.4 getSize()

```
int Inventory::getSize ( ) const
```

6.14.1.5 isEmpty()

```
bool Inventory::isEmpty ( ) const
```

6.14.1.6 isFull()

```
bool Inventory::isFull ( ) const
```

6.14.1.7 newSize()

```
void Inventory::newSize ( )
```

6.14.1.8 operator[]()

```
const Item * Inventory::operator[] (
    int index ) const
```

6.14.1.9 removeBrokenItems()

```
void Inventory::removeBrokenItems ( )
```

6.14.1.10 removeItem()

```
void Inventory::removeItem (
    int index )
```

6.14.1.11 useItem()

```
void Inventory::useItem (
    int index,
    Character & user,
    Character * enemy = nullptr )
```

6.14.2 Member Data Documentation

6.14.2.1 inventorySize

```
int Inventory::inventorySize { 5 } [private]
```

6.14.2.2 m_items

```
std::vector<std::unique_ptr<Item> > Inventory::m_items [private]
```

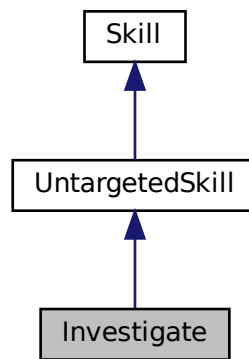
The documentation for this class was generated from the following files:

- src/Equipment/[Inventory.h](#)
- src/Equipment/[Inventory.cpp](#)

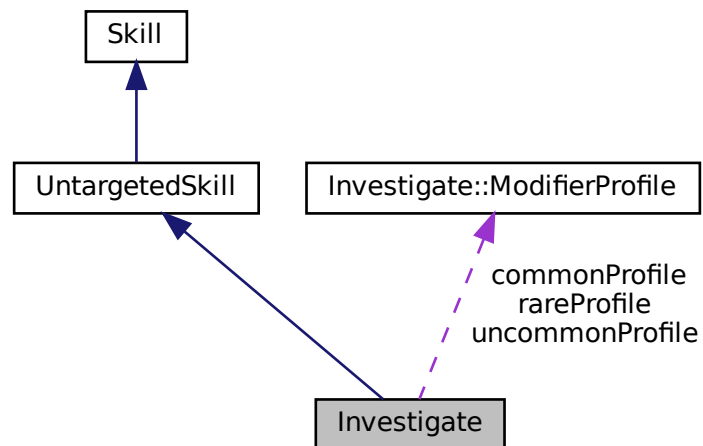
6.15 Investigate Class Reference

```
#include <Investigate.h>
```

Inheritance diagram for Investigate:



Collaboration diagram for Investigate:



Classes

- struct [ModifierProfile](#)

Public Member Functions

- [Investigate](#) ()
- void [execute](#) ([Character](#) &user) override
- std::string_view [getName](#) () const override

Static Public Member Functions

- static std::unique_ptr< [Item](#) > [createGenericItem](#) ()

Static Private Member Functions

- static std::unique_ptr< [Item](#) > [createEpicItem](#) ()
- static [StatModifier](#) [generateModifier](#) (const [ModifierProfile](#) &profile)
- static std::string [getFriendlyNameForSingle](#) ([Stat](#) stat, int bonus, [Target](#) target)
- static std::string [generateMultiModifierName](#) ([Rarity](#) overallRarity, const std::vector< [StatModifier](#) > &mods)
- static std::vector< [StatModifier](#) > [combineModifiers](#) (const std::vector< [StatModifier](#) > &mods)
- static bool [isSumValid](#) ([Rarity](#) rarity, int totalSum)

Static Private Attributes

- static const [ModifierProfile](#) [commonProfile](#)
- static const [ModifierProfile](#) [uncommonProfile](#)
- static const [ModifierProfile](#) [rareProfile](#)

Additional Inherited Members

6.15.1 Constructor & Destructor Documentation

6.15.1.1 Investigate()

```
Investigate::Investigate ( ) [inline]
```

6.15.2 Member Function Documentation

6.15.2.1 combineModifiers()

```
std::vector< StatModifier > Investigate::combineModifiers (
    const std::vector< StatModifier > & mods ) [static], [private]
```

6.15.2.2 createEpicItem()

```
std::unique_ptr< Item > Investigate::createEpicItem ( ) [static], [private]
```

6.15.2.3 createGenericItem()

```
std::unique_ptr< Item > Investigate::createGenericItem ( ) [static]
```

6.15.2.4 execute()

```
void Investigate::execute (
    Character & user ) [override], [virtual]
```

Implements [UntargetedSkill](#).

6.15.2.5 generateModifier()

```
StatModifier Investigate::generateModifier (
    const ModifierProfile & profile ) [static], [private]
```

6.15.2.6 generateMultiModifierName()

```
std::string Investigate::generateMultiModifierName (
    Rarity overallRarity,
    const std::vector< StatModifier > & mods ) [static], [private]
```

6.15.2.7 getFriendlyNameForSingle()

```
std::string Investigate::getFriendlyNameForSingle (
    Stat stat,
    int bonus,
    Target target ) [static], [private]
```

6.15.2.8 getName()

```
std::string_view Investigate::getName ( ) const [inline], [override], [virtual]
```

Implements [Skill](#).

6.15.2.9 isSumValid()

```
bool Investigate::isSumValid (
    Rarity rarity,
    int totalSum ) [static], [private]
```

6.15.3 Member Data Documentation

6.15.3.1 commonProfile

```
const Investigate::ModifierProfile Investigate::commonProfile [static], [private]
```

Initial value:

```
{
    1, 2, 5,
    100, 5, 10,
    0, 10, 20,
    0, 30, 40
}
```

6.15.3.2 rareProfile

```
const Investigate::ModifierProfile Investigate::rareProfile [static], [private]
```

Initial value:

```
{
    1, 2, 0,
    20, 5, 10,
    60, 10, 20,
    40, 30, 40
}
```

6.15.3.3 uncommonProfile

```
const Investigate::ModifierProfile Investigate::uncommonProfile [static], [private]
```

Initial value:

```
{
    1, 3, 10,
    70, 5, 10,
    30, 10, 20,
    0, 30, 40
}
```

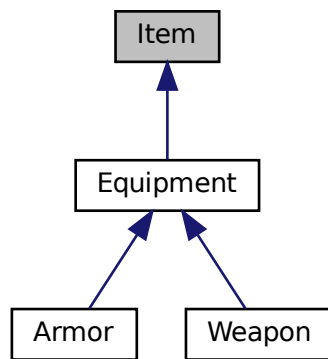
The documentation for this class was generated from the following files:

- [src/Skills/Investigate.h](#)
- [src/Skills/Investigate.cpp](#)

6.16 Item Class Reference

```
#include <Item.h>
```

Inheritance diagram for Item:



Public Member Functions

- [Item](#) (std::string_view name, [Rarity](#) rarity, const std::vector< [StatModifier](#) > &modifiers, std::function< void([Character](#) &)> customEffect=nullptr)
- [Item](#) (std::string_view name, [Rarity](#) rarity, std::function< void([Character](#) &)> customEffect=nullptr)
- virtual [~Item](#) ()=default
- const std::string & [getName](#) () const
- [Rarity](#) [getRarity](#) () const
- const std::vector< [StatModifier](#) > & [getModifiers](#) () const
- std::string_view [getRarityName](#) () const
- void [setCustomEffect](#) (std::function< void([Character](#) &)> effect)
- void [runCustomEffect](#) ([Character](#) &user)
- void [use](#) ([Character](#) &user, [Character](#) *[enemy](#)=nullptr)

Protected Attributes

- std::string [m_name](#)
- [Rarity](#) [m_rarity](#)
- std::vector< [StatModifier](#) > [m_modifiers](#)
- std::function< void([Character](#) &)> [m_customEffect](#)
- bool [m_effectApplied](#) { false }

6.16.1 Constructor & Destructor Documentation

6.16.1.1 Item() [1/2]

```
Item::Item (
    std::string_view name,
    Rarity rarity,
    const std::vector< StatModifier > & modifiers,
    std::function< void(Character &)> customEffect = nullptr )
```

6.16.1.2 Item() [2/2]

```
Item::Item (
    std::string_view name,
    Rarity rarity,
    std::function< void(Character &)> customEffect = nullptr )
```

6.16.1.3 ~Item()

```
virtual Item::~Item ( ) [virtual], [default]
```

6.16.2 Member Function Documentation

6.16.2.1 getModifiers()

```
const std::vector<StatModifier>& Item::getModifiers ( ) const [inline]
```

6.16.2.2 getName()

```
const std::string& Item::getName ( ) const [inline]
```

6.16.2.3 getRarity()

```
Rarity Item::getRarity ( ) const [inline]
```

6.16.2.4 getRarityName()

```
std::string_view Item::getRarityName ( ) const
```

6.16.2.5 runCustomEffect()

```
void Item::runCustomEffect (
    Character & user )
```

6.16.2.6 setCustomEffect()

```
void Item::setCustomEffect (
    std::function< void(Character &)> effect ) [inline]
```

6.16.2.7 use()

```
void Item::use (
    Character & user,
    Character * enemy = nullptr )
```

6.16.3 Member Data Documentation

6.16.3.1 m_customEffect

```
std::function<void(Character&)> Item::m_customEffect [protected]
```

6.16.3.2 m_effectApplied

```
bool Item::m_effectApplied { false } [protected]
```

6.16.3.3 m_modifiers

```
std::vector<StatModifier> Item::m_modifiers [protected]
```

6.16.3.4 m_name

```
std::string Item::m_name [protected]
```

6.16.3.5 m_rarity

```
Rarity Item::m_rarity [protected]
```

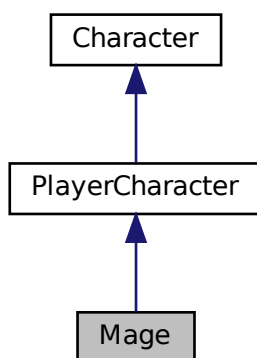
The documentation for this class was generated from the following files:

- src/Equipment/[Item.h](#)
- src/Equipment/[Item.cpp](#)

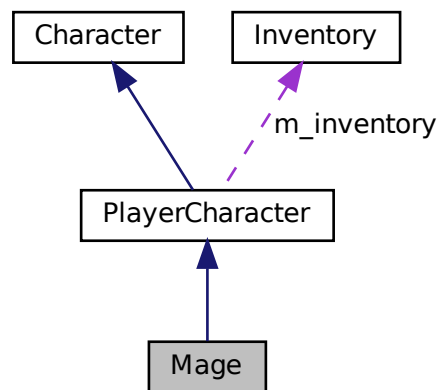
6.17 Mage Class Reference

```
#include <Mage.h>
```

Inheritance diagram for Mage:



Collaboration diagram for Mage:



Public Member Functions

- [Mage](#) (std::string_view name)
- void [startNewTurn](#) ()
- int [getTurnCounter](#) () const

Private Attributes

- int [m_turnCounter](#) { 0 }

Additional Inherited Members

6.17.1 Constructor & Destructor Documentation

6.17.1.1 Mage()

```
Mage::Mage (
    std::string_view name )
```

6.17.2 Member Function Documentation

6.17.2.1 getTurnCounter()

```
int Mage::getTurnCounter ( ) const [inline]
```

6.17.2.2 startNewTurn()

```
void Mage::startNewTurn ( )
```

6.17.3 Member Data Documentation

6.17.3.1 m_turnCounter

```
int Mage::m_turnCounter { 0 } [private]
```

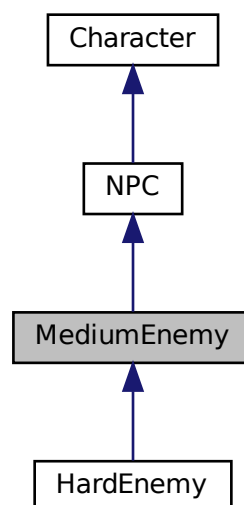
The documentation for this class was generated from the following files:

- [src/Character/Mage.h](#)
- [src/Character/Mage.cpp](#)

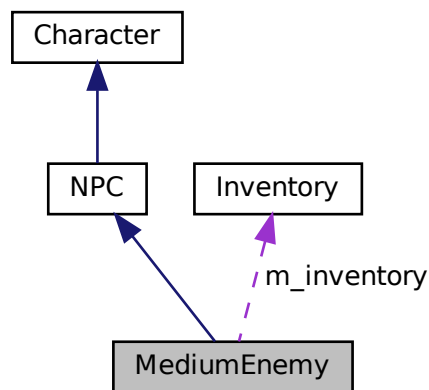
6.18 MediumEnemy Class Reference

```
#include <MediumEnemy.h>
```

Inheritance diagram for MediumEnemy:



Collaboration diagram for MediumEnemy:



Public Member Functions

- [MediumEnemy](#) (std::string_view name="Sofiya", int [strength](#)=6, int [intelligence](#)=6, int [agility](#)=7, int [mana](#)=50, int [health](#)=300, int AP=8)
- virtual [~MediumEnemy](#) ()=default
- void [performTurn](#) ([PlayerCharacter](#) &player) override

Protected Member Functions

- void [equipBackup](#) () override
- bool [shouldUseUltra](#) (double playerHealthRatio, double enemyHealthRatio) override
- void [performUseItem](#) ([PlayerCharacter](#) &player, int bestIndex)
- std::pair< int, int > [evaluateInventory](#) () const
- int [evaluateItem](#) (const [Item](#) &item) const

Protected Attributes

- [Inventory](#) [m_inventory](#)
- int [m_ultraAttacksUsed](#)
- int [m_turnsSinceUltra](#)

6.18.1 Constructor & Destructor Documentation

6.18.1.1 MediumEnemy()

```
MediumEnemy::MediumEnemy (
    std::string_view name = "Sofiya",
    int strength = 6,
    int intelligence = 6,
    int agility = 7,
    int mana = 50,
    int health = 300,
    int AP = 8 )
```

6.18.1.2 ~MediumEnemy()

```
virtual MediumEnemy::~~MediumEnemy ( ) [virtual], [default]
```

6.18.2 Member Function Documentation

6.18.2.1 equipBackup()

```
void MediumEnemy::equipBackup ( ) [override], [protected], [virtual]
```

Implements [NPC](#).

6.18.2.2 evaluateInventory()

```
std::pair< int, int > MediumEnemy::evaluateInventory ( ) const [protected]
```

6.18.2.3 evaluateItem()

```
int MediumEnemy::evaluateItem (
    const Item & item ) const [protected]
```

6.18.2.4 performTurn()

```
void MediumEnemy::performTurn (
    PlayerCharacter & player ) [override], [virtual]
```

Implements [NPC](#).

6.18.2.5 performUseItem()

```
void MediumEnemy::performUseItem (
    PlayerCharacter & player,
    int bestIndex )    [protected]
```

6.18.2.6 shouldUseUltra()

```
bool MediumEnemy::shouldUseUltra (
    double playerHealthRatio,
    double enemyHealthRatio )    [override], [protected], [virtual]
```

Reimplemented from [NPC](#).

6.18.3 Member Data Documentation

6.18.3.1 m_inventory

```
Inventory MediumEnemy::m_inventory    [protected]
```

6.18.3.2 m_turnsSinceUltra

```
int MediumEnemy::m_turnsSinceUltra    [protected]
```

6.18.3.3 m_ultraAttacksUsed

```
int MediumEnemy::m_ultraAttacksUsed    [protected]
```

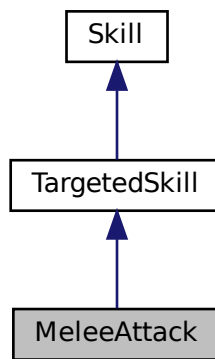
The documentation for this class was generated from the following files:

- [src/Character/MediumEnemy.h](#)
- [src/Character/MediumEnemy.cpp](#)

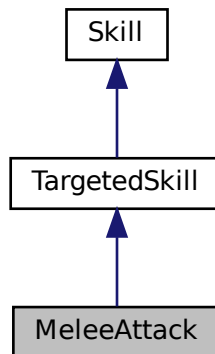
6.19 MeleeAttack Class Reference

```
#include <MeleeAttack.h>
```

Inheritance diagram for MeleeAttack:



Collaboration diagram for MeleeAttack:



Public Member Functions

- [MeleeAttack](#) (int baseDamage=10, int cost=[Constants::melee_cost](#))
- void [execute](#) ([Character](#) &user, [Character](#) &target) override
- std::string_view [getName](#) () const override

Additional Inherited Members

6.19.1 Constructor & Destructor Documentation

6.19.1.1 MeleeAttack()

```
MeleeAttack::MeleeAttack (
    int baseDamage = 10,
    int cost = Constants::melee_cost )
```

6.19.2 Member Function Documentation

6.19.2.1 execute()

```
void MeleeAttack::execute (
    Character & user,
    Character & target ) [override], [virtual]
```

Implements [TargetedSkill](#).

6.19.2.2 getName()

```
std::string_view MeleeAttack::getName ( ) const [inline], [override], [virtual]
```

Implements [Skill](#).

The documentation for this class was generated from the following files:

- [src/Skills/MeleeAttack.h](#)
- [src/Skills/MeleeAttack.cpp](#)

6.20 Investigate::ModifierProfile Struct Reference

Public Attributes

- int [minMods](#)
- int [maxMods](#)
- int [negativeChance](#)
- int [weakChance](#)
- int [weakMin](#)
- int [weakMax](#)
- int [mediocreChance](#)
- int [mediocreMin](#)
- int [mediocreMax](#)
- int [strongChance](#)
- int [strongMin](#)
- int [strongMax](#)

6.20.1 Member Data Documentation

6.20.1.1 maxMods

```
int Investigate::ModifierProfile::maxMods
```

6.20.1.2 mediocreChance

```
int Investigate::ModifierProfile::mediocreChance
```

6.20.1.3 mediocreMax

```
int Investigate::ModifierProfile::mediocreMax
```

6.20.1.4 mediocreMin

```
int Investigate::ModifierProfile::mediocreMin
```

6.20.1.5 minMods

```
int Investigate::ModifierProfile::minMods
```

6.20.1.6 negativeChance

```
int Investigate::ModifierProfile::negativeChance
```

6.20.1.7 strongChance

```
int Investigate::ModifierProfile::strongChance
```

6.20.1.8 strongMax

```
int Investigate::ModifierProfile::strongMax
```

6.20.1.9 strongMin

```
int Investigate::ModifierProfile::strongMin
```

6.20.1.10 weakChance

```
int Investigate::ModifierProfile::weakChance
```

6.20.1.11 weakMax

```
int Investigate::ModifierProfile::weakMax
```

6.20.1.12 weakMin

```
int Investigate::ModifierProfile::weakMin
```

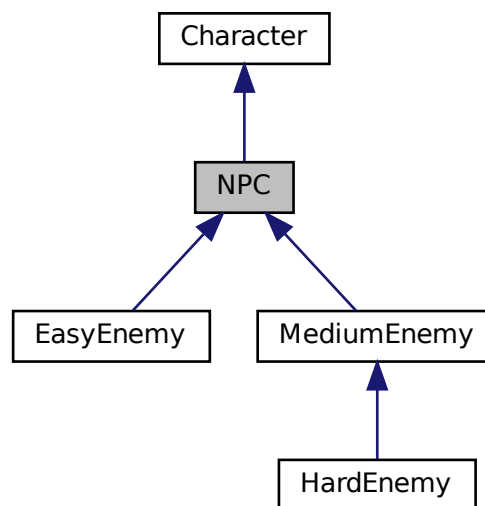
The documentation for this struct was generated from the following file:

- [src/Skills/Investigate.h](#)

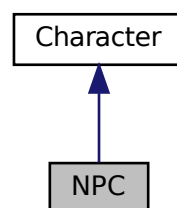
6.21 NPC Class Reference

```
#include <NPC.h>
```

Inheritance diagram for NPC:



Collaboration diagram for NPC:



Public Member Functions

- **NPC** (std::string_view name, int strength, int intelligence, int agility, int mana, int health, int AP)
- virtual **~NPC** ()=default
- virtual void **performTurn** (PlayerCharacter &player)=0
- void **displayStatus** () const override

Protected Member Functions

- virtual void [equipBackup](#) ()=0
- virtual bool [shouldUseUltra](#) (double playerHealthRatio, double enemyHealthRatio)
- bool [shouldHeal](#) (double playerHealthRatio, double enemyHealthRatio)
- bool [shouldRestoreMana](#) ()
- bool [shouldBuff](#) ()
- bool [shouldSpellCast](#) ()
- void [performUltraAttack](#) (PlayerCharacter &player)
- void [performMeleeAttack](#) (PlayerCharacter &player)
- void [performSpellAttack](#) (PlayerCharacter &player)
- void [performHeal](#) ()
- void [performBuff](#) ()
- void [performRestoreMana](#) ()

Protected Attributes

- bool [m_hasBuff](#)
- bool [m_hasEquippedBackup](#) {}

6.21.1 Constructor & Destructor Documentation

6.21.1.1 NPC()

```
NPC::NPC (
    std::string_view name,
    int strength,
    int intelligence,
    int agility,
    int mana,
    int health,
    int AP )
```

6.21.1.2 ~NPC()

```
virtual NPC::~NPC ( ) [virtual], [default]
```

6.21.2 Member Function Documentation

6.21.2.1 displayStatus()

```
void NPC::displayStatus ( ) const [override], [virtual]
```

Implements [Character](#).

6.21.2.2 equipBackup()

```
virtual void NPC::equipBackup ( ) [protected], [pure virtual]
```

Implemented in [MediumEnemy](#), [HardEnemy](#), and [EasyEnemy](#).

6.21.2.3 performBuff()

```
void NPC::performBuff ( ) [protected]
```

6.21.2.4 performHeal()

```
void NPC::performHeal ( ) [protected]
```

6.21.2.5 performMeleeAttack()

```
void NPC::performMeleeAttack (
    PlayerCharacter & player ) [protected]
```

6.21.2.6 performRestoreMana()

```
void NPC::performRestoreMana ( ) [protected]
```

6.21.2.7 performSpellAttack()

```
void NPC::performSpellAttack (
    PlayerCharacter & player ) [protected]
```

6.21.2.8 performTurn()

```
virtual void NPC::performTurn (
    PlayerCharacter & player ) [pure virtual]
```

Implemented in [MediumEnemy](#), [HardEnemy](#), and [EasyEnemy](#).

6.21.2.9 performUltraAttack()

```
void NPC::performUltraAttack (
    PlayerCharacter & player ) [protected]
```

6.21.2.10 shouldBuff()

```
bool NPC::shouldBuff ( ) [protected]
```

6.21.2.11 shouldHeal()

```
bool NPC::shouldHeal (
    double playerHealthRatio,
    double enemyHealthRatio ) [protected]
```

6.21.2.12 shouldRestoreMana()

```
bool NPC::shouldRestoreMana ( ) [protected]
```

6.21.2.13 shouldSpellCast()

```
bool NPC::shouldSpellCast ( ) [protected]
```

6.21.2.14 shouldUseUltra()

```
bool NPC::shouldUseUltra (
    double playerHealthRatio,
    double enemyHealthRatio ) [protected], [virtual]
```

Reimplemented in [MediumEnemy](#).

6.21.3 Member Data Documentation

6.21.3.1 m_hasBuff

```
bool NPC::m_hasBuff [protected]
```

6.21.3.2 m_hasEquippedBackup

```
bool NPC::m_hasEquippedBackup {} [protected]
```

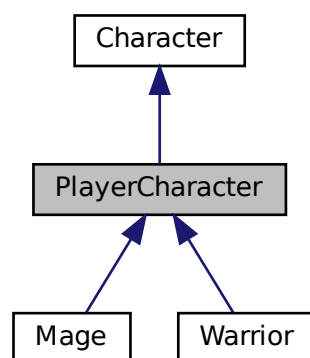
The documentation for this class was generated from the following files:

- [src/Character/NPC.h](#)
- [src/Character/NPC.cpp](#)

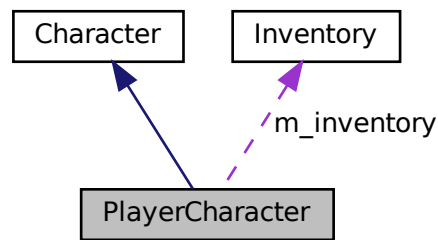
6.22 PlayerCharacter Class Reference

```
#include <PlayerCharacter.h>
```

Inheritance diagram for PlayerCharacter:



Collaboration diagram for PlayerCharacter:



Public Member Functions

- [PlayerCharacter](#) ()
- [PlayerCharacter](#) (std::string_view name, int [agility](#), int [mana](#))
- void [displayStatus](#) () const override
- [Inventory](#) & [getInventory](#) ()
- const [Inventory](#) & [getInventory](#) () const
- void [enableSpellCast](#) ()
- void [unlockHealing](#) ()
- void [boostConcentration](#) ()
- void [reduceBlockCost](#) ()
- bool [canUseSpell](#) () const
- bool [hasHealing](#) () const
- bool [hasMonkTraining](#) () const
- bool [hasEasyBlock](#) () const
- void [updateArmorBonuses](#) ()
- int [getInvestigateCost](#) () const
- int [getStrengthBonus](#) () const

Private Attributes

- [Inventory](#) [m_inventory](#)
- bool [hasSpellCastSkill](#) { false }
- bool [canHeal](#) { false }
- bool [monkTraining](#) { false }
- bool [easyBlock](#) { false }
- int [investigateCost](#) { [Constants::invest_cost](#) }
- int [strengthBonus](#) { 0 }

Additional Inherited Members

6.22.1 Constructor & Destructor Documentation

6.22.1.1 PlayerCharacter() [1/2]

```
PlayerCharacter::PlayerCharacter ( ) [inline]
```

6.22.1.2 PlayerCharacter() [2/2]

```
PlayerCharacter::PlayerCharacter (
    std::string_view name,
    int agility,
    int mana ) [inline]
```

6.22.2 Member Function Documentation

6.22.2.1 boostConcentration()

```
void PlayerCharacter::boostConcentration ( ) [inline]
```

6.22.2.2 canUseSpell()

```
bool PlayerCharacter::canUseSpell ( ) const [inline]
```

6.22.2.3 displayStatus()

```
void PlayerCharacter::displayStatus ( ) const [override], [virtual]
```

Implements [Character](#).

6.22.2.4 enableSpellCast()

```
void PlayerCharacter::enableSpellCast ( ) [inline]
```

6.22.2.5 `getInventory()` [1/2]

```
Inventory& PlayerCharacter::getInventory ( ) [inline]
```

6.22.2.6 `getInventory()` [2/2]

```
const Inventory& PlayerCharacter::getInventory ( ) const [inline]
```

6.22.2.7 `getInvestigateCost()`

```
int PlayerCharacter::getInvestigateCost ( ) const [inline]
```

6.22.2.8 `getStrengthBonus()`

```
int PlayerCharacter::getStrengthBonus ( ) const [inline]
```

6.22.2.9 `hasEasyBlock()`

```
bool PlayerCharacter::hasEasyBlock ( ) const [inline]
```

6.22.2.10 `hasHealing()`

```
bool PlayerCharacter::hasHealing ( ) const [inline]
```

6.22.2.11 `hasMonkTraining()`

```
bool PlayerCharacter::hasMonkTraining ( ) const [inline]
```

6.22.2.12 `reduceBlockCost()`

```
void PlayerCharacter::reduceBlockCost ( ) [inline]
```

6.22.2.13 unlockHealing()

```
void PlayerCharacter::unlockHealing ( ) [inline]
```

6.22.2.14 updateArmorBonuses()

```
void PlayerCharacter::updateArmorBonuses ( )
```

6.22.3 Member Data Documentation

6.22.3.1 canHeal

```
bool PlayerCharacter::canHeal { false } [private]
```

6.22.3.2 easyBlock

```
bool PlayerCharacter::easyBlock { false } [private]
```

6.22.3.3 hasSpellCastSkill

```
bool PlayerCharacter::hasSpellCastSkill { false } [private]
```

6.22.3.4 investigateCost

```
int PlayerCharacter::investigateCost { Constants::invest_cost } [private]
```

6.22.3.5 m_inventory

```
Inventory PlayerCharacter::m_inventory [private]
```

6.22.3.6 monkTraining

```
bool PlayerCharacter::monkTraining { false } [private]
```

6.22.3.7 strengthBonus

```
int PlayerCharacter::strengthBonus { 0 } [private]
```

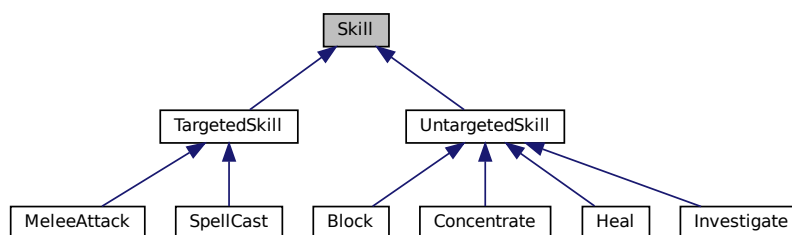
The documentation for this class was generated from the following files:

- src/Character/[PlayerCharacter.h](#)
- src/Character/[PlayerCharacter.cpp](#)

6.23 Skill Class Reference

```
#include <Skill.h>
```

Inheritance diagram for Skill:



Public Types

- enum [Type](#) { [Damage](#) , [Heal](#) , [Defense](#) , [Utility](#) }

Public Member Functions

- [Skill](#) ([Type](#) type, int cost=0)
- virtual [~Skill](#) ()=default
- int [getCost](#) () const
- virtual std::string_view [getName](#) () const =0

Protected Attributes

- [Type](#) m_type
- int m_cost

6.23.1 Member Enumeration Documentation

6.23.1.1 Type

```
enum Skill::Type
```

Enumerator

Damage	
Heal	
Defense	
Utility	

6.23.2 Constructor & Destructor Documentation

6.23.2.1 Skill()

```
Skill::Skill (
    Type type,
    int cost = 0 )
```

6.23.2.2 ~Skill()

```
virtual Skill::~Skill ( ) [virtual], [default]
```

6.23.3 Member Function Documentation

6.23.3.1 getCost()

```
int Skill::getCost ( ) const [inline]
```

6.23.3.2 getName()

```
virtual std::string_view Skill::getName ( ) const [pure virtual]
```

Implemented in [SpellCast](#), [MeleeAttack](#), [Investigate](#), [Heal](#), [Concentrate](#), and [Block](#).

6.23.4 Member Data Documentation

6.23.4.1 m_cost

```
int Skill::m_cost [protected]
```

6.23.4.2 m_type

```
Type Skill::m_type [protected]
```

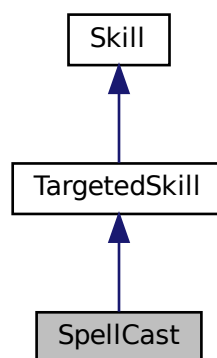
The documentation for this class was generated from the following files:

- [src/Skills/Skill.h](#)
- [src/Skills/Skill.cpp](#)

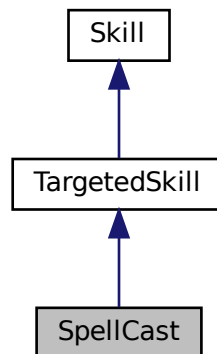
6.24 SpellCast Class Reference

```
#include <SpellCast.h>
```

Inheritance diagram for SpellCast:



Collaboration diagram for SpellCast:



Public Member Functions

- [SpellCast](#) (int manaCost=15, int [damage](#)=15, int cost=[Constants::cast_cost](#))
- void [execute](#) ([Character](#) &user, [Character](#) &target) override
- std::string_view [getName](#) () const override

Private Attributes

- int [m_manaCost](#)

Additional Inherited Members

6.24.1 Constructor & Destructor Documentation

6.24.1.1 SpellCast()

```
SpellCast::SpellCast (  
    int manaCost = 15,  
    int damage = 15,  
    int cost = Constants::cast\_cost )
```

6.24.2 Member Function Documentation

6.24.2.1 execute()

```
void SpellCast::execute (
    Character & user,
    Character & target ) [override], [virtual]
```

Implements [TargetedSkill](#).

6.24.2.2 getName()

```
std::string_view SpellCast::getName ( ) const [inline], [override], [virtual]
```

Implements [Skill](#).

6.24.3 Member Data Documentation

6.24.3.1 m_manacost

```
int SpellCast::m_manacost [private]
```

The documentation for this class was generated from the following files:

- [src/Skills/SpellCast.h](#)
- [src/Skills/SpellCast.cpp](#)

6.25 StatModifier Struct Reference

```
#include <StatModifier.h>
```

Public Attributes

- [Stat](#) stat {}
- int [value](#) {}
- [Target](#) target { [Target::self](#) }

6.25.1 Member Data Documentation

6.25.1.1 stat

```
Stat StatModifier::stat {}
```

6.25.1.2 target

```
Target StatModifier::target { Target::self }
```

6.25.1.3 value

```
int StatModifier::value {}
```

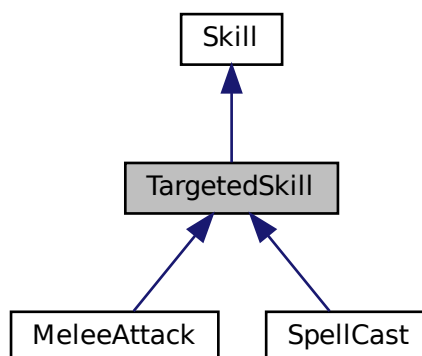
The documentation for this struct was generated from the following file:

- src/Equipment/[StatModifier.h](#)

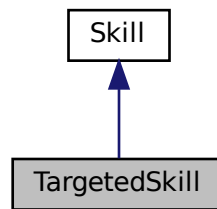
6.26 TargetedSkill Class Reference

```
#include <TargetedSkill.h>
```

Inheritance diagram for TargetedSkill:



Collaboration diagram for TargetedSkill:



Public Member Functions

- [TargetedSkill](#) ([Type](#) type, int cost, int [damage](#))
- int [getDamage](#) ()
- virtual void [execute](#) ([Character](#) &user, [Character](#) &target)=0

Private Attributes

- int [m_damage](#)

Additional Inherited Members

6.26.1 Constructor & Destructor Documentation

6.26.1.1 TargetedSkill()

```
TargetedSkill::TargetedSkill (  
    Type type,  
    int cost,  
    int damage ) [inline]
```

6.26.2 Member Function Documentation

6.26.2.1 execute()

```
virtual void TargetedSkill::execute (
    Character & user,
    Character & target ) [pure virtual]
```

Implemented in [SpellCast](#), and [MeleeAttack](#).

6.26.2.2 getDamage()

```
int TargetedSkill::getDamage ( ) [inline]
```

6.26.3 Member Data Documentation

6.26.3.1 m_damage

```
int TargetedSkill::m_damage [private]
```

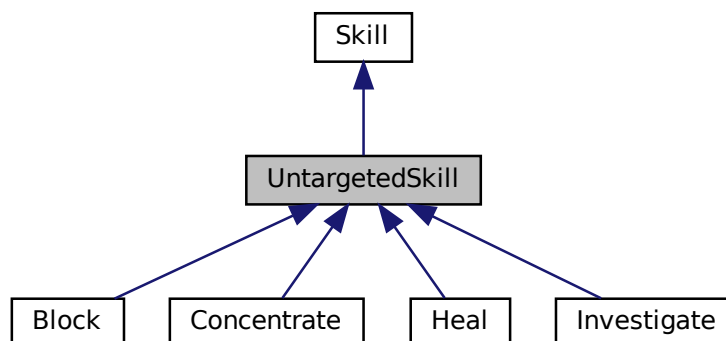
The documentation for this class was generated from the following file:

- [src/Skills/TargetedSkill.h](#)

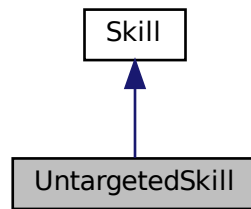
6.27 UntargetedSkill Class Reference

```
#include <UntargetedSkill.h>
```

Inheritance diagram for UntargetedSkill:



Collaboration diagram for UntargetedSkill:



Public Member Functions

- [UntargetedSkill](#) ([Type](#) type, int cost)
- virtual void [execute](#) ([Character](#) &user)=0

Additional Inherited Members

6.27.1 Constructor & Destructor Documentation

6.27.1.1 UntargetedSkill()

```
UntargetedSkill::UntargetedSkill (
    Type type,
    int cost ) [inline]
```

6.27.2 Member Function Documentation

6.27.2.1 execute()

```
virtual void UntargetedSkill::execute (
    Character & user ) [pure virtual]
```

Implemented in [Investigate](#), [Heal](#), [Concentrate](#), and [Block](#).

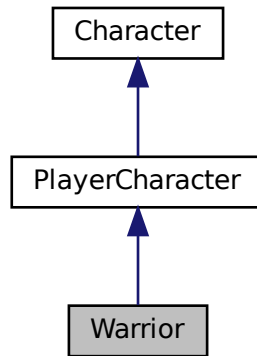
The documentation for this class was generated from the following file:

- [src/Skills/UntargetedSkill.h](#)

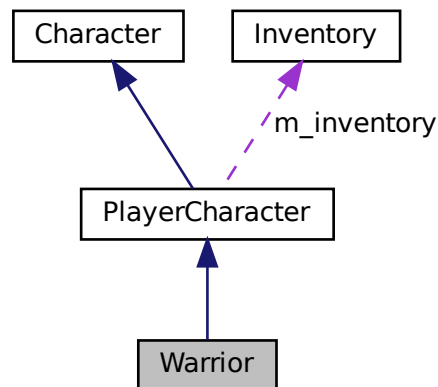
6.28 Warrior Class Reference

```
#include <Warrior.h>
```

Inheritance diagram for Warrior:



Collaboration diagram for Warrior:



Public Member Functions

- [Warrior](#) (std::string_view name)
- void [setBlockSuccess](#) (bool success)
- bool [isDamageBuffActive](#) () const
- void [resetDamageBuff](#) ()
- void [resetActionPoints](#) () override

Private Attributes

- bool `m_blockSuccess` { false }
- bool `m_damageBuff` { false }

Additional Inherited Members

6.28.1 Constructor & Destructor Documentation

6.28.1.1 Warrior()

```
Warrior::Warrior (
    std::string_view name )
```

6.28.2 Member Function Documentation

6.28.2.1 isDamageBuffActive()

```
bool Warrior::isDamageBuffActive ( ) const
```

6.28.2.2 resetActionPoints()

```
void Warrior::resetActionPoints ( ) [override], [virtual]
```

Reimplemented from [Character](#).

6.28.2.3 resetDamageBuff()

```
void Warrior::resetDamageBuff ( )
```

6.28.2.4 setBlockSuccess()

```
void Warrior::setBlockSuccess (
    bool success )
```

6.28.3 Member Data Documentation

6.28.3.1 m_blockSuccess

```
bool Warrior::m_blockSuccess { false } [private]
```

6.28.3.2 m_damageBuff

```
bool Warrior::m_damageBuff { false } [private]
```

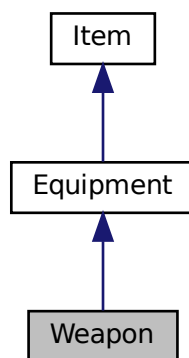
The documentation for this class was generated from the following files:

- src/Character/[Warrior.h](#)
- src/Character/[Warrior.cpp](#)

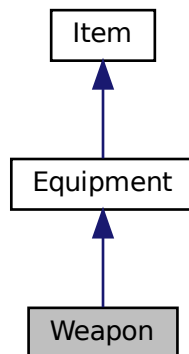
6.29 Weapon Class Reference

```
#include <Weapon.h>
```

Inheritance diagram for Weapon:



Collaboration diagram for Weapon:



Public Member Functions

- [Weapon](#) (std::string_view name, int [damage](#), int durability)
- void [use](#) ([Character](#) &user, std::unique_ptr< [Item](#) > [self](#)) override
- int [getBaseDamage](#) () const

Private Attributes

- int [baseDamage](#)

Additional Inherited Members

6.29.1 Constructor & Destructor Documentation

6.29.1.1 Weapon()

```
Weapon::Weapon (
    std::string_view name,
    int damage,
    int durability )
```

6.29.2 Member Function Documentation

6.29.2.1 `getBaseDamage()`

```
int Weapon::getBaseDamage ( ) const
```

6.29.2.2 `use()`

```
void Weapon::use (
    Character & user,
    std::unique_ptr< Item > self ) [override], [virtual]
```

Implements [Equipment](#).

6.29.3 Member Data Documentation

6.29.3.1 `baseDamage`

```
int Weapon::baseDamage [private]
```

The documentation for this class was generated from the following files:

- [src/Equipment/Weapon.h](#)
- [src/Equipment/Weapon.cpp](#)

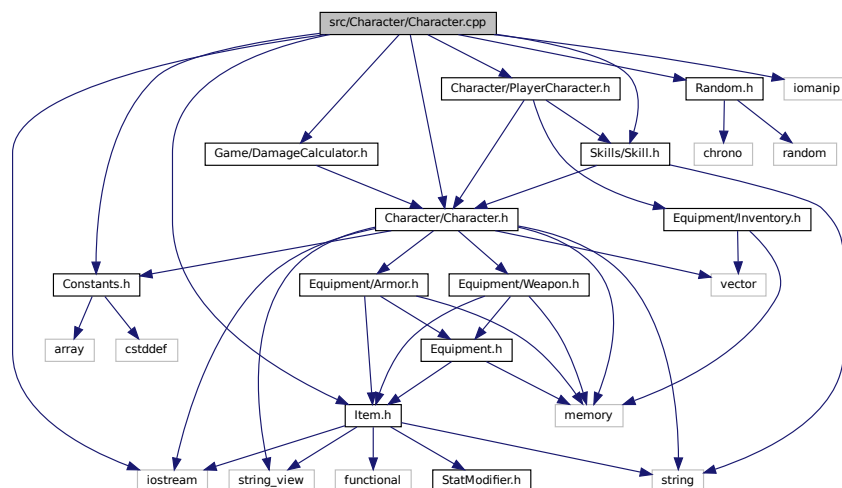
Chapter 7

File Documentation

7.1 src/Character/Character.cpp File Reference

```
#include "Character/Character.h"  
#include "Character/PlayerCharacter.h"  
#include "Skills/Skill.h"  
#include "Equipment/Item.h"  
#include "Game/DamageCalculator.h"  
#include "Constants.h"  
#include "Random.h"  
#include <iostream>  
#include <iomanip>
```

Include dependency graph for Character.cpp:

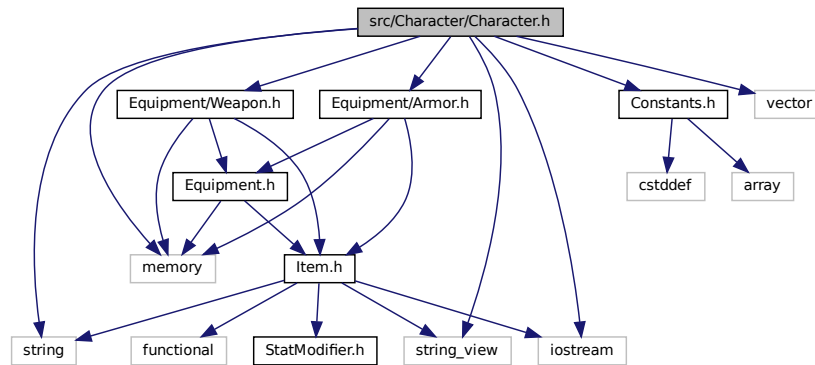


7.2 src/Character/Character.h File Reference

```
#include "Equipment/Weapon.h"  
#include "Equipment/Armor.h"
```

```
#include "Constants.h"
#include <string>
#include <string_view>
#include <vector>
#include <memory>
#include <iostream>
```

Include dependency graph for Character.h:



This graph shows which files directly or indirectly include this file:



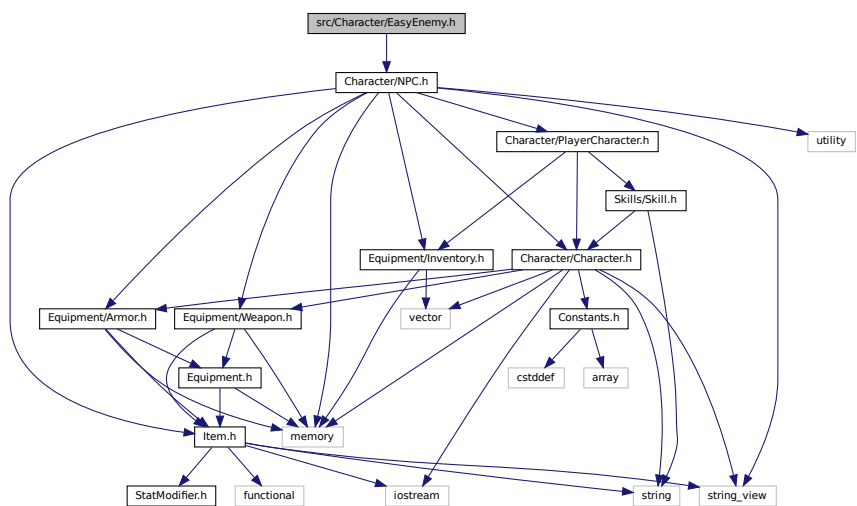
Classes

- class [Character](#)

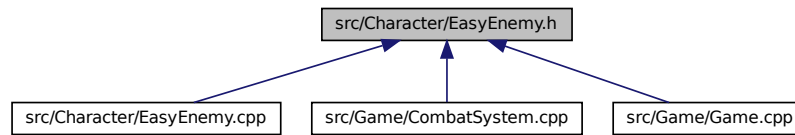
7.3 src/Character/EasyEnemy.cpp File Reference

```
#include "EasyEnemy.h"
#include <iostream>
#include <algorithm>
```

```
#include "Character/NPC.h"
Include dependency graph for EasyEnemy.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [EasyEnemy](#)

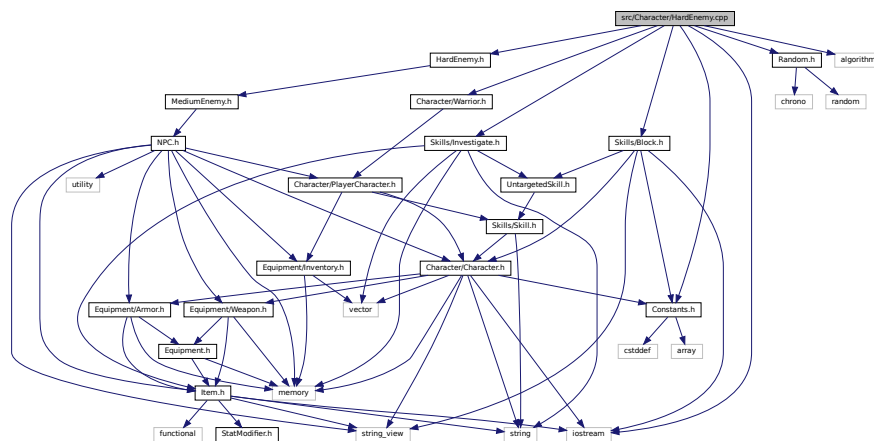
7.5 src/Character/HardEnemy.cpp File Reference

```

#include "HardEnemy.h"
#include "Skills/Block.h"
#include "Skills/Investigate.h"
#include "Character/Warrior.h"
#include "Constants.h"
#include "Random.h"
#include <iostream>
#include <algorithm>

```

Include dependency graph for HardEnemy.cpp:



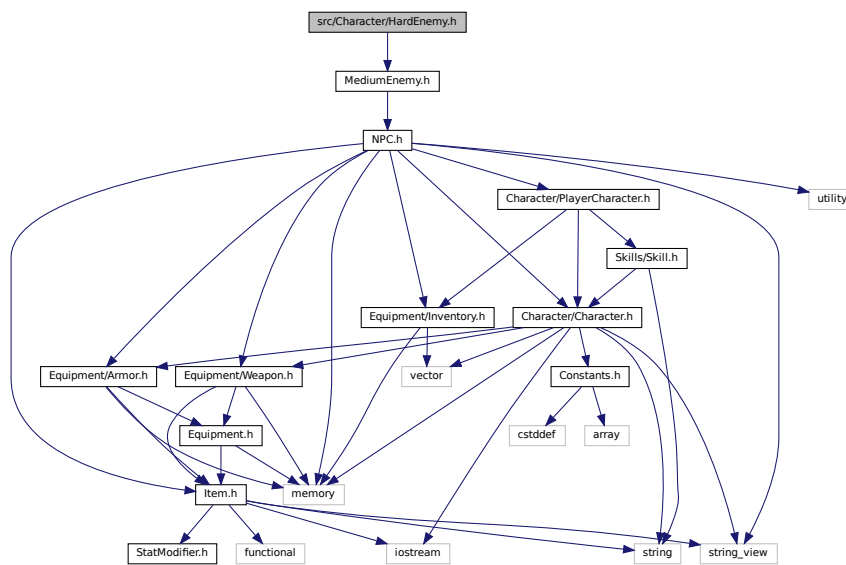
7.6 src/Character/HardEnemy.h File Reference

```

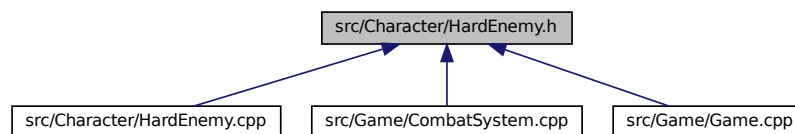
#include "MediumEnemy.h"

```


Include dependency graph for HardEnemy.h:



This graph shows which files directly or indirectly include this file:



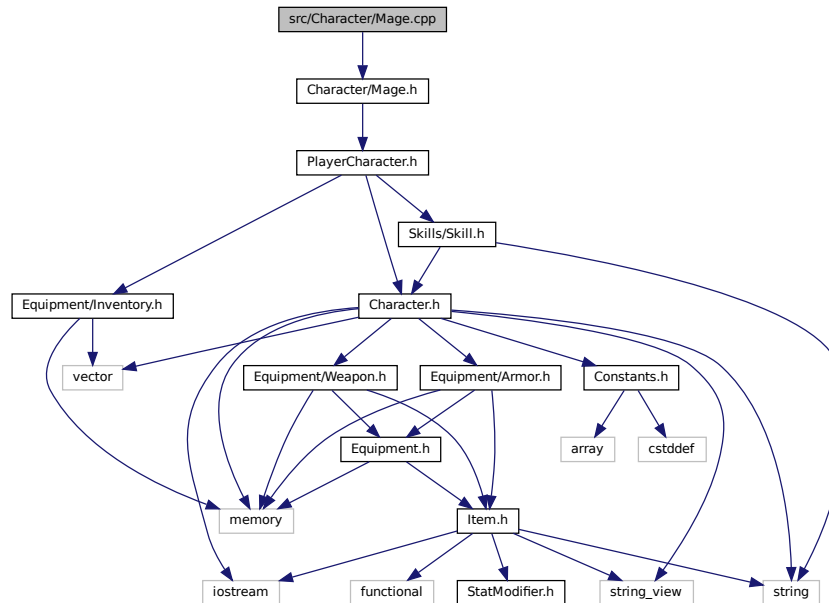
Classes

- class [HardEnemy](#)

7.7 src/Character/Mage.cpp File Reference

```
#include "Character/Mage.h"
```

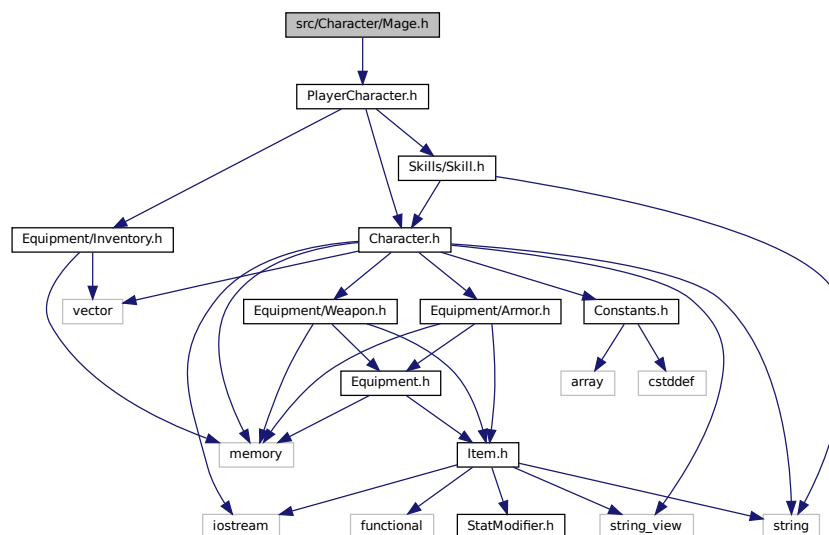
Include dependency graph for Mage.cpp:



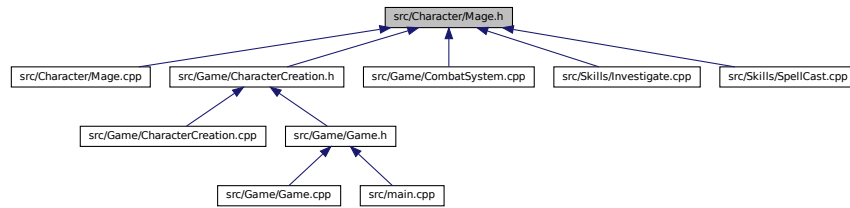
7.8 src/Character/Mage.h File Reference

```
#include "PlayerCharacter.h"
```

Include dependency graph for Mage.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [Mage](#)

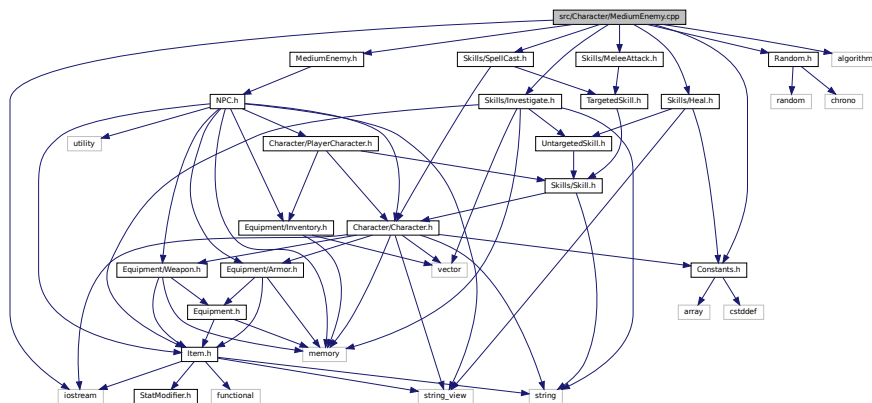
7.9 src/Character/MediumEnemy.cpp File Reference

```

#include "MediumEnemy.h"
#include "Skills/Investigate.h"
#include "Skills/MeleeAttack.h"
#include "Skills/SpellCast.h"
#include "Skills/Heal.h"
#include "Random.h"
#include "Constants.h"
#include <iostream>
#include <algorithm>

```

Include dependency graph for MediumEnemy.cpp:



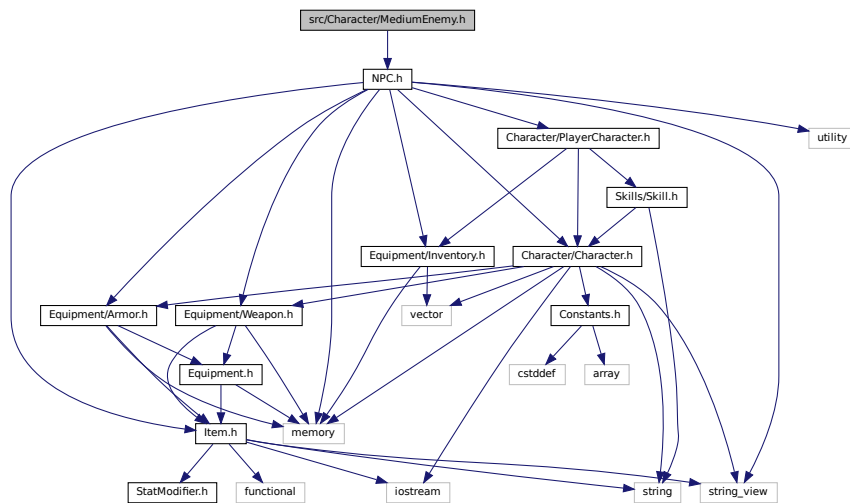
7.10 src/Character/MediumEnemy.h File Reference

```

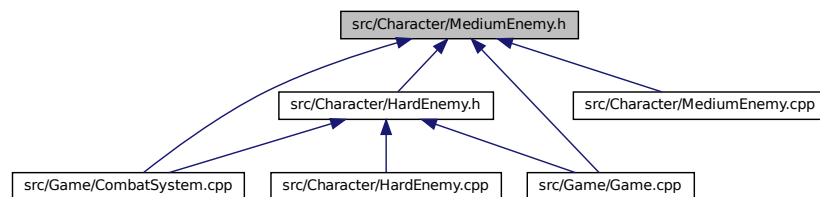
#include "NPC.h"

```

Include dependency graph for MediumEnemy.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [MediumEnemy](#)

7.11 src/Character/NPC.cpp File Reference

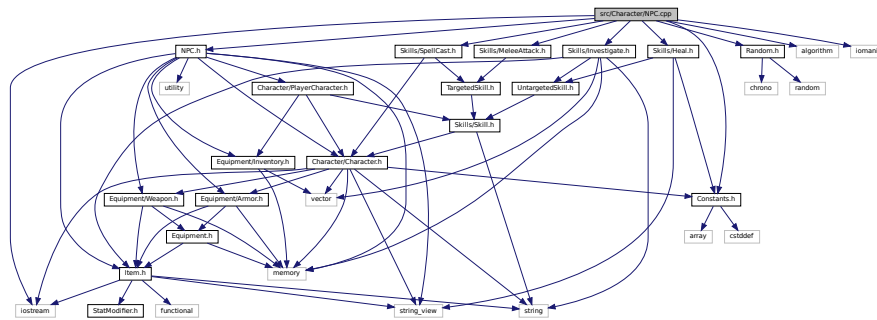
```

#include "NPC.h"
#include "Skills/Investigate.h"
#include "Skills/MeleeAttack.h"
#include "Skills/SpellCast.h"
#include "Skills/Heal.h"
#include "Random.h"
#include "Constants.h"
#include <iostream>
#include <algorithm>

```

```
#include <iomanip>
```

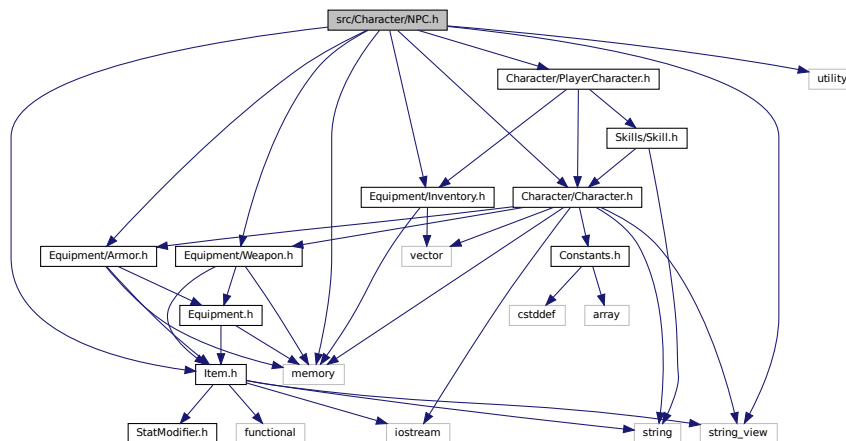
Include dependency graph for NPC.cpp:



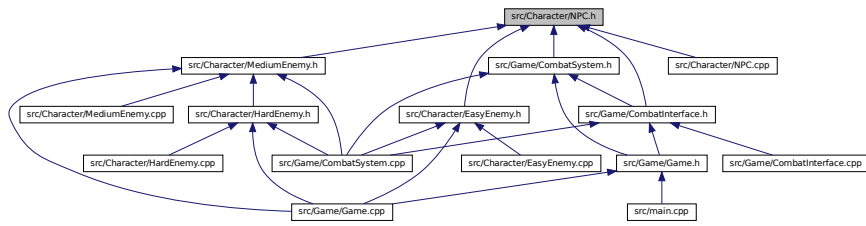
7.12 src/Character/NPC.h File Reference

```
#include "Character/Character.h"
#include "Equipment/Weapon.h"
#include "Equipment/Armor.h"
#include "Character/PlayerCharacter.h"
#include "Equipment/Inventory.h"
#include "Equipment/Item.h"
#include <memory>
#include <utility>
#include <string_view>
```

Include dependency graph for NPC.h:



This graph shows which files directly or indirectly include this file:



Classes

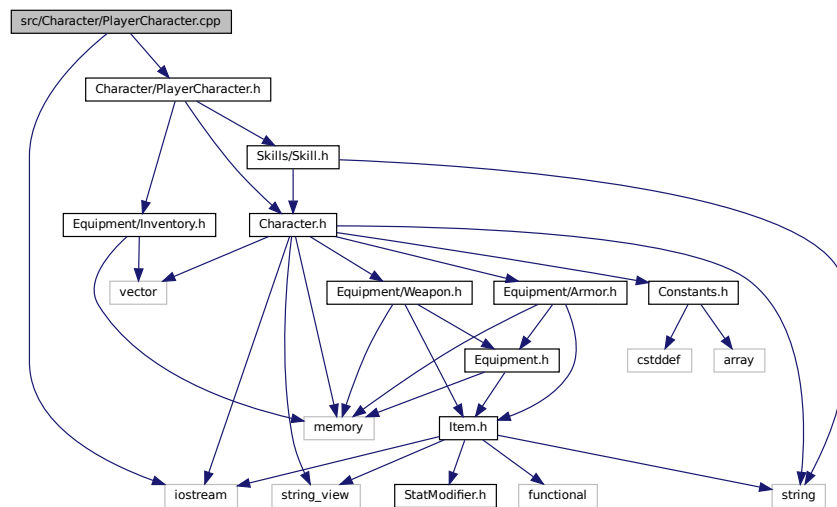
- class [NPC](#)

7.13 src/Character/PlayerCharacter.cpp File Reference

```
#include "Character/PlayerCharacter.h"
```

```
#include <iostream>
```

Include dependency graph for PlayerCharacter.cpp:

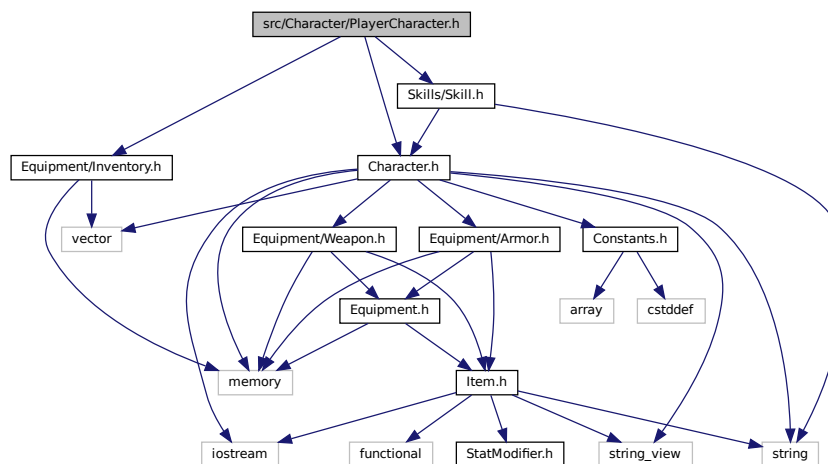


7.14 src/Character/PlayerCharacter.h File Reference

```
#include "Character.h"
```

```
#include "Equipment/Inventory.h"
```

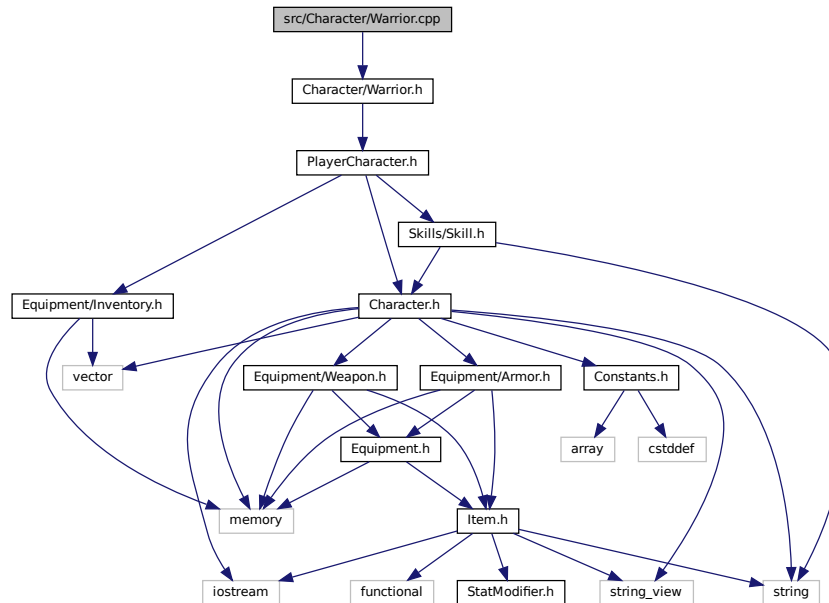
Include dependency graph for PlayerCharacter.h:

[illegible]

- class **PlayerCharacter**

```
#include "Character/Warrior.h"
```

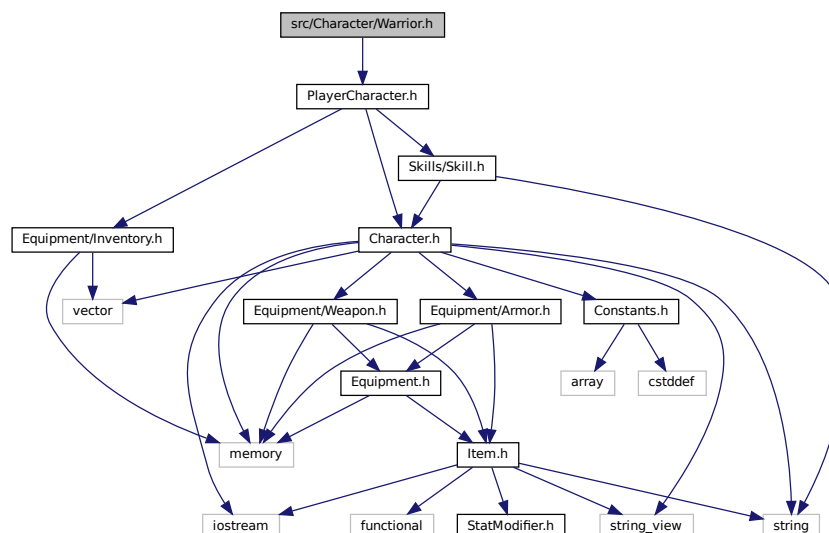
Include dependency graph for Warrior.cpp:



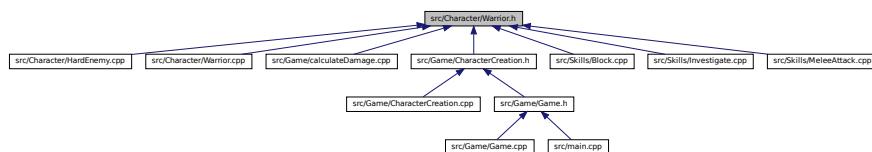
7.16 src/Character/Warrior.h File Reference

```
#include "PlayerCharacter.h"
```

Include dependency graph for Warrior.h:



This graph shows which files directly or indirectly include this file:



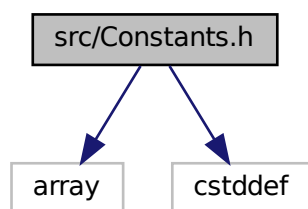
Classes

- class [Warrior](#)

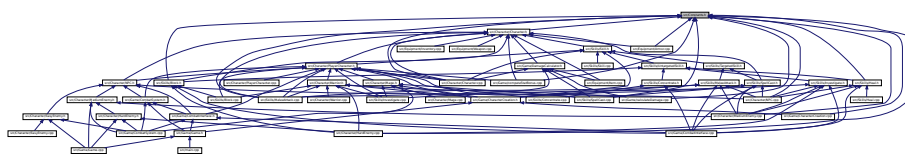
7.17 src/Constants.h File Reference

```
#include <array>
#include <cstdint>
```

Include dependency graph for Constants.h:



This graph shows which files directly or indirectly include this file:



Namespaces

- [Constants](#)

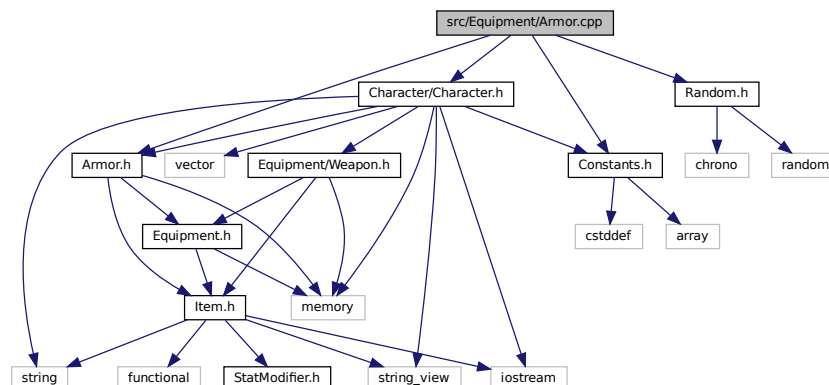
Variables

- constexpr int [Constants::min_attack_bonus](#) { 1 }
- constexpr int [Constants::max_attack_bonus](#) { 5 }
- constexpr int [Constants::min_crit_multiplier](#) { 200 }
- constexpr int [Constants::max_crit_multiplier](#) { 299 }
- constexpr int [Constants::critThreshold](#) { 3 }
- constexpr std::array [Constants::statThreshold](#) { 6, 8, 10, 13, 16, 20 }
- constexpr double [Constants::low_bonus](#) { -0.1 }
- constexpr double [Constants::high_bonus](#) { 0.2 }
- constexpr int [Constants::min_armor_reduce](#) { 40 }
- constexpr int [Constants::max_armor_reduce](#) { 60 }
- constexpr int [Constants::min_block_reduce](#) { 50 }
- constexpr int [Constants::max_block_reduce](#) { 100 }
- constexpr int [Constants::min_weapon_decoy](#) { 2 }
- constexpr int [Constants::max_weapon_decoy](#) { 5 }
- constexpr int [Constants::points_to_distribute](#) { 10 }
- constexpr int [Constants::initial_ap](#) { 10 }
- constexpr int [Constants::concentrationManaRestore](#) { 10 }
- constexpr int [Constants::healAmount](#) { 20 }
- constexpr size_t [Constants::maxInventorySize](#) { 5 }
- constexpr int [Constants::block_cost](#) { 5 }
- constexpr int [Constants::melee_cost](#) { 5 }
- constexpr int [Constants::cast_cost](#) { 3 }
- constexpr int [Constants::invest_cost](#) { 2 }
- constexpr int [Constants::conc_cost](#) { 5 }
- constexpr int [Constants::heal_cost](#) { 2 }

7.18 src/Equipment/Armor.cpp File Reference

```
#include "Armor.h"
#include "Character/Character.h"
#include "Random.h"
#include "Constants.h"
```

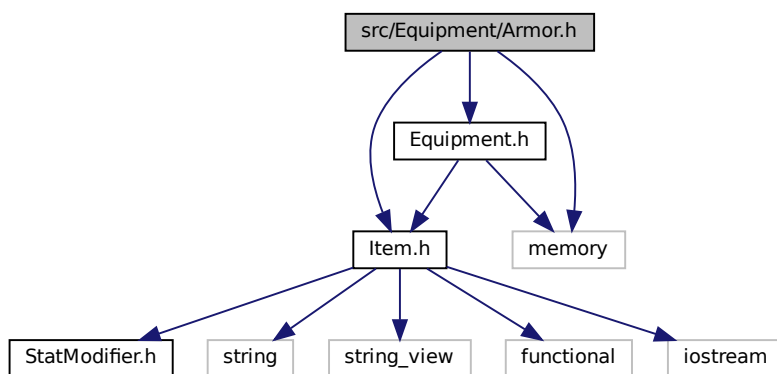
Include dependency graph for Armor.cpp:



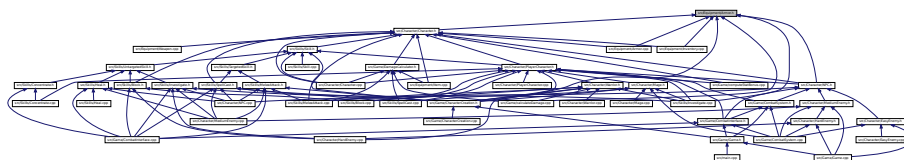
7.19 src/Equipment/Armor.h File Reference

```
#include "Item.h"
#include "Equipment.h"
#include <memory>
```

Include dependency graph for Armor.h:



This graph shows which files directly or indirectly include this file:



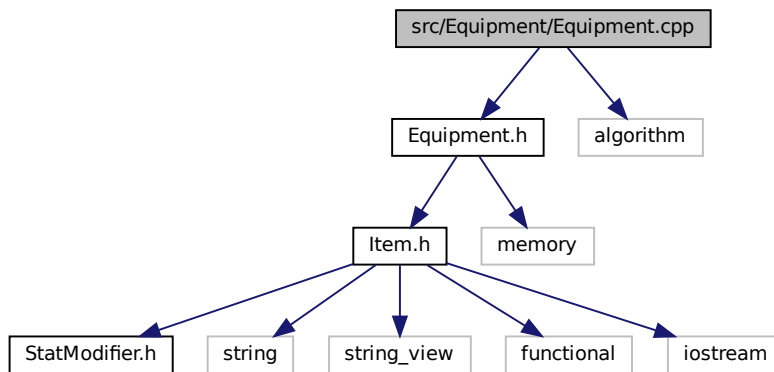
Classes

- class [Armor](#)

7.20 src/Equipment/Equipment.cpp File Reference

```
#include "Equipment.h"
#include <algorithm>
```

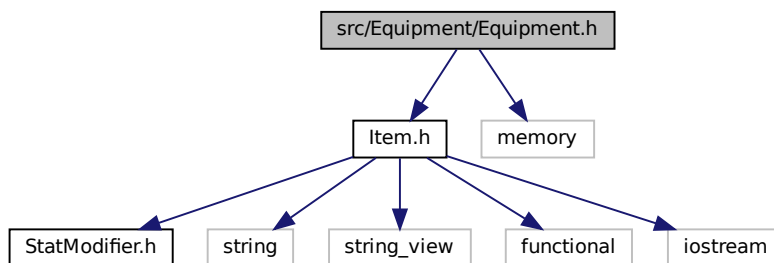
Include dependency graph for Equipment.cpp:



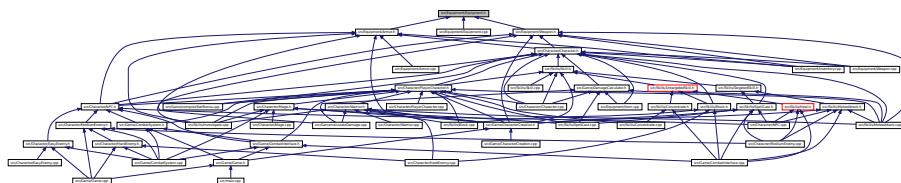
7.21 src/Equipment/Equipment.h File Reference

```
#include "Item.h"
#include <memory>
```

Include dependency graph for Equipment.h:



This graph shows which files directly or indirectly include this file:



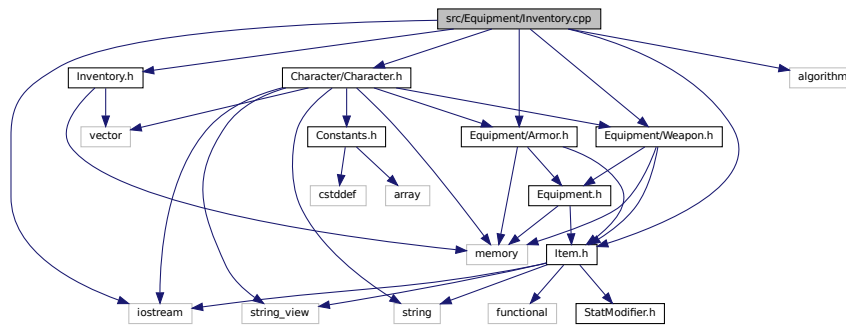
Classes

- class [Equipment](#)

7.22 src/Equipment/Inventory.cpp File Reference

```
#include "Inventory.h"
#include "Item.h"
#include "Character/Character.h"
#include "Equipment/Weapon.h"
#include "Equipment/Armor.h"
#include <iostream>
#include <algorithm>
```

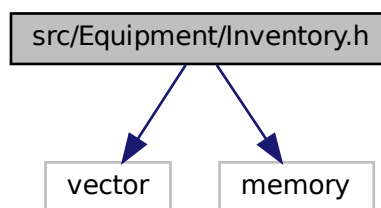
Include dependency graph for Inventory.cpp:



7.23 src/Equipment/Inventory.h File Reference

```
#include <vector>
#include <memory>
```

Include dependency graph for Inventory.h:



This graph shows which files directly or indirectly include this file:

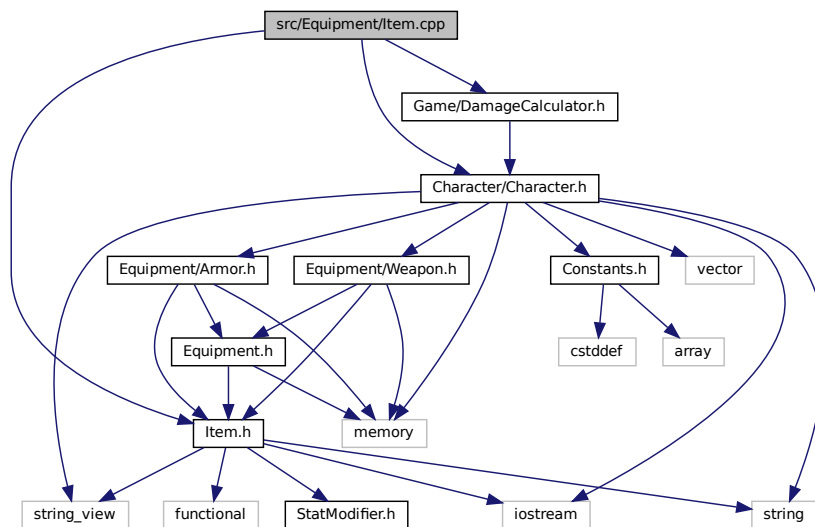


Classes

- class [Inventory](#)

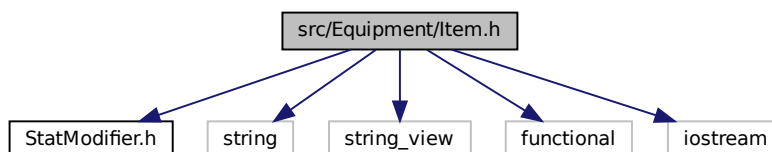
7.24 src/Equipment/Item.cpp File Reference

```
#include "Item.h"
#include "Game/DamageCalculator.h"
#include "Character/Character.h"
Include dependency graph for Item.cpp:
```

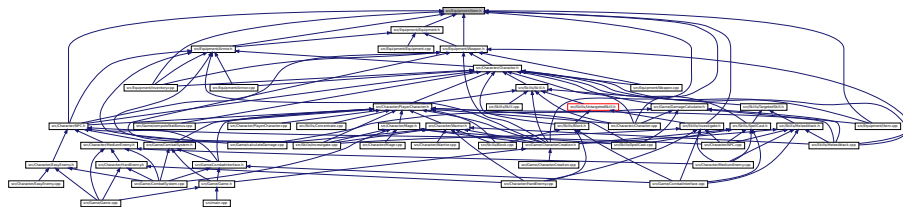


7.25 src/Equipment/Item.h File Reference

```
#include "StatModifier.h"
#include <string>
#include <string_view>
#include <functional>
#include <iostream>
Include dependency graph for Item.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [Item](#)

Enumerations

- enum class [Rarity](#) { [common](#) , [uncommon](#) , [rare](#) , [epic](#) }

7.25.1 Enumeration Type Documentation

7.25.1.1 Rarity

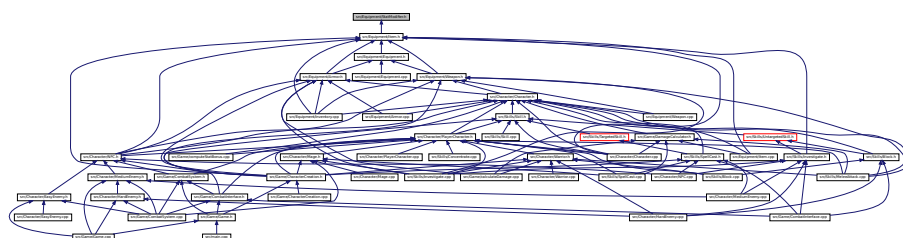
```
enum Rarity [strong]
```

Enumerator

common	
uncommon	
rare	
epic	

7.26 src/Equipment/StatModifier.h File Reference

This graph shows which files directly or indirectly include this file:



Classes

- struct [StatModifier](#)

Enumerations

- enum class [Target](#) { [self](#) , [enemy](#) , [armor](#) , [weapon](#) }
- enum class [Stat](#) {
 [none](#) , [health](#) , [mana](#) , [maxHealth](#) ,
 [maxMana](#) , [strength](#) , [intelligence](#) , [agility](#) ,
 [actionPoints](#) , [repair](#) , [damage](#) }

7.26.1 Enumeration Type Documentation

7.26.1.1 Stat

```
enum Stat [strong]
```

Enumerator

none	
health	
mana	
maxHealth	
maxMana	
strength	
intelligence	
agility	
actionPoints	
repair	
damage	

7.26.1.2 Target

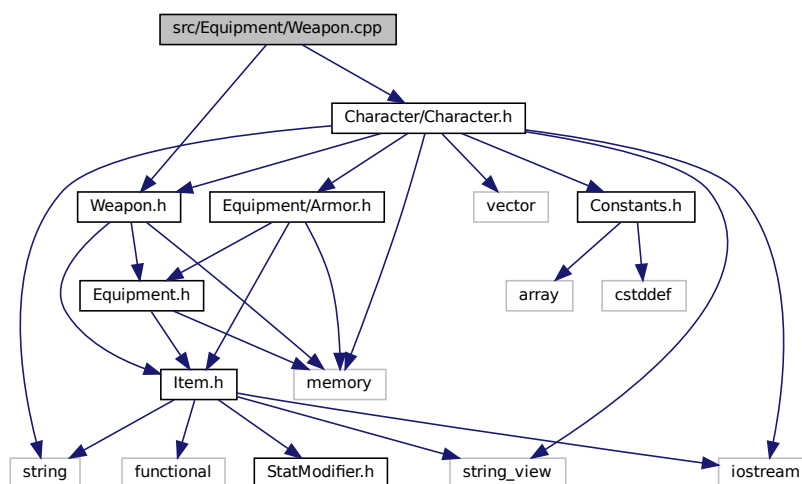
```
enum Target [strong]
```

Enumerator

self	
enemy	
armor	
weapon	

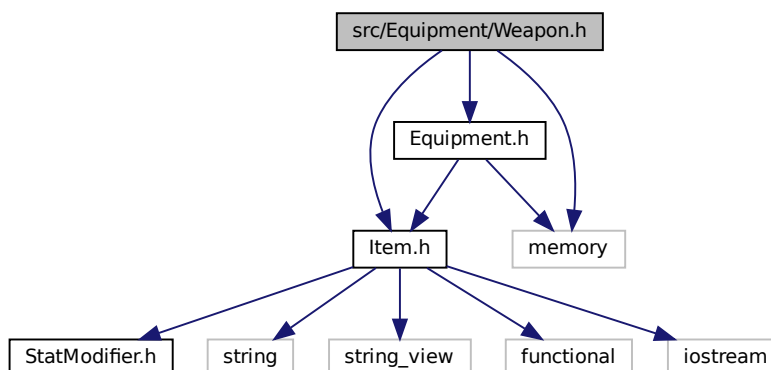
7.27 src/Equipment/Weapon.cpp File Reference

```
#include "Weapon.h"
#include "Character/Character.h"
Include dependency graph for Weapon.cpp:
```

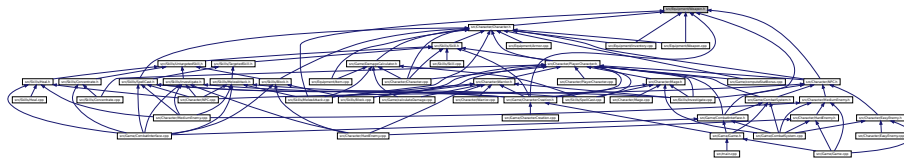


7.28 src/Equipment/Weapon.h File Reference

```
#include "Item.h"
#include "Equipment.h"
#include <memory>
Include dependency graph for Weapon.h:
```



This graph shows which files directly or indirectly include this file:



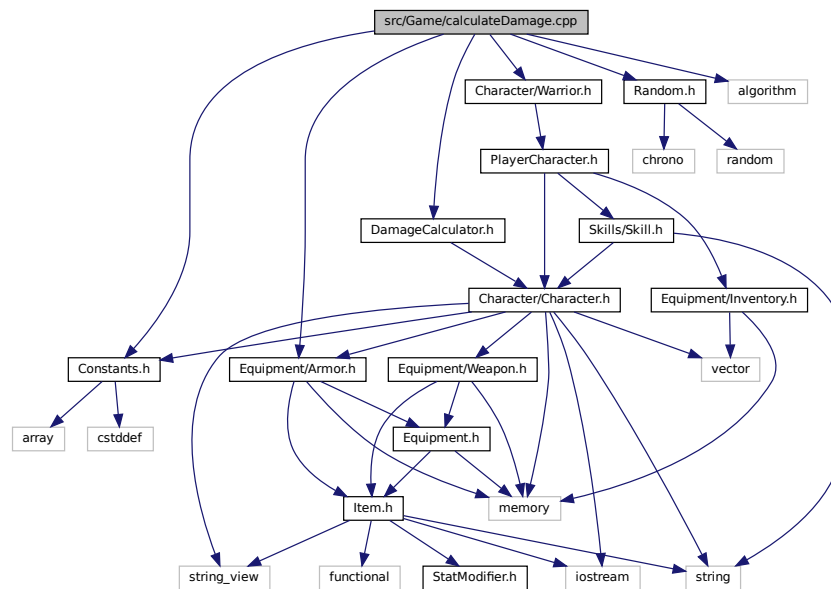
Classes

- class [Weapon](#)

7.29 src/Game/calculateDamage.cpp File Reference

```
#include "DamageCalculator.h"
#include "Equipment/Armor.h"
#include "Character/Warrior.h"
#include "Random.h"
#include "Constants.h"
#include <algorithm>
```

Include dependency graph for calculateDamage.cpp:



Functions

- void [calculateDamage](#) ([DamageCalculator](#) &dmgCalc, [Character](#) &target, int baseDamage, int statBonus, int attackCount)

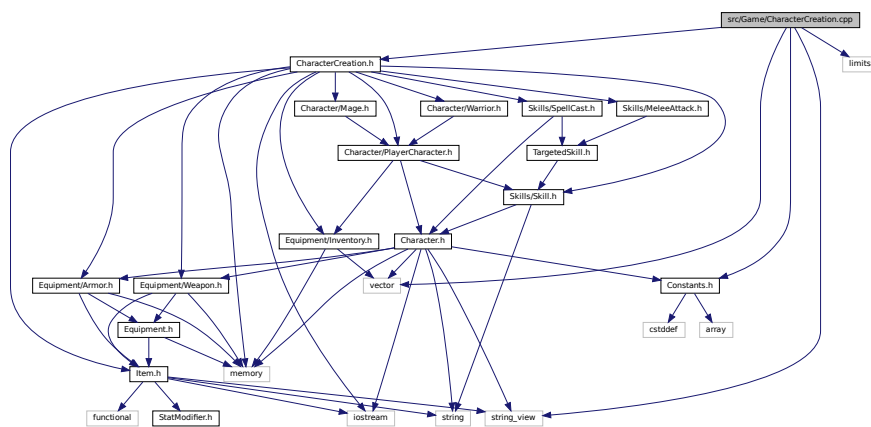
7.29.1 Function Documentation

7.29.1.1 calculateDamage()

```
void calculateDamage (
    DamageCalculator & dmgCalc,
    Character & target,
    int baseDamage,
    int statBonus,
    int attackCount )
```

7.30 src/Game/CharacterCreation.cpp File Reference

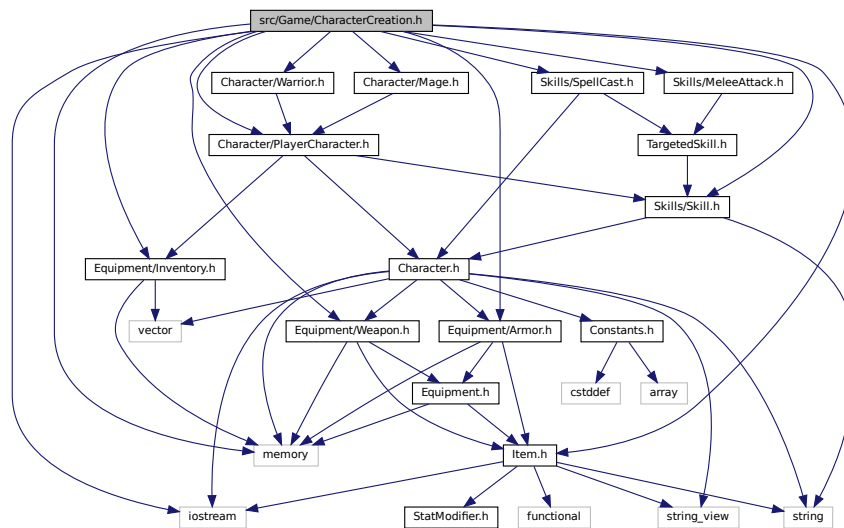
```
#include "CharacterCreation.h"
#include "Constants.h"
#include <limits>
#include <vector>
#include <string_view>
Include dependency graph for CharacterCreation.cpp:
```



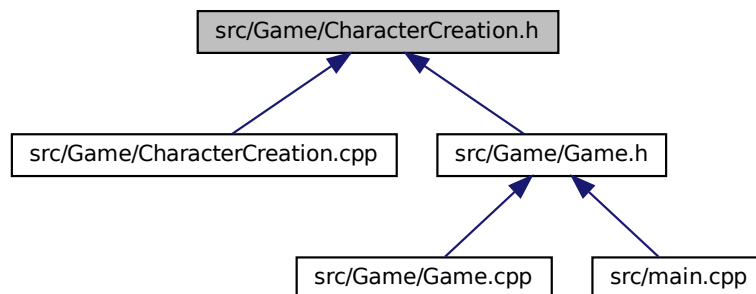
7.31 src/Game/CharacterCreation.h File Reference

```
#include "Character/PlayerCharacter.h"
#include "Character/Warrior.h"
#include "Character/Mage.h"
#include "Equipment/Weapon.h"
#include "Equipment/Armor.h"
#include "Skills/Skill.h"
#include "Skills/MeleeAttack.h"
#include "Skills/SpellCast.h"
#include "Equipment/Inventory.h"
#include "Equipment/Item.h"
```

```
#include <iostream>
#include <memory>
Include dependency graph for CharacterCreation.h:
```



This graph shows which files directly or indirectly include this file:



Classes

- class [CharacterCreation](#)

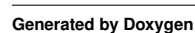
7.32 src/Game/CombatInterface.cpp File Reference

```
#include "CombatInterface.h"
#include "Equipment/Inventory.h"
#include "Skills/Block.h"
#include "Skills/Heal.h"
```

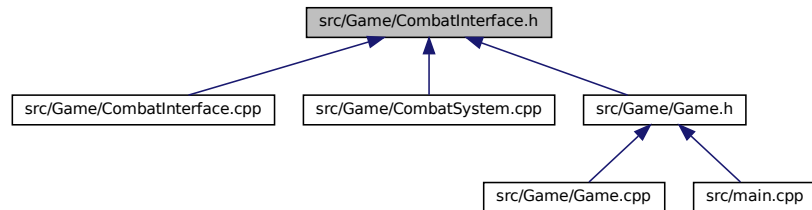
Include dependency graph for CombatInterface.cpp:



Include dependency graph for CombatInterface.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [CombatInterface](#)

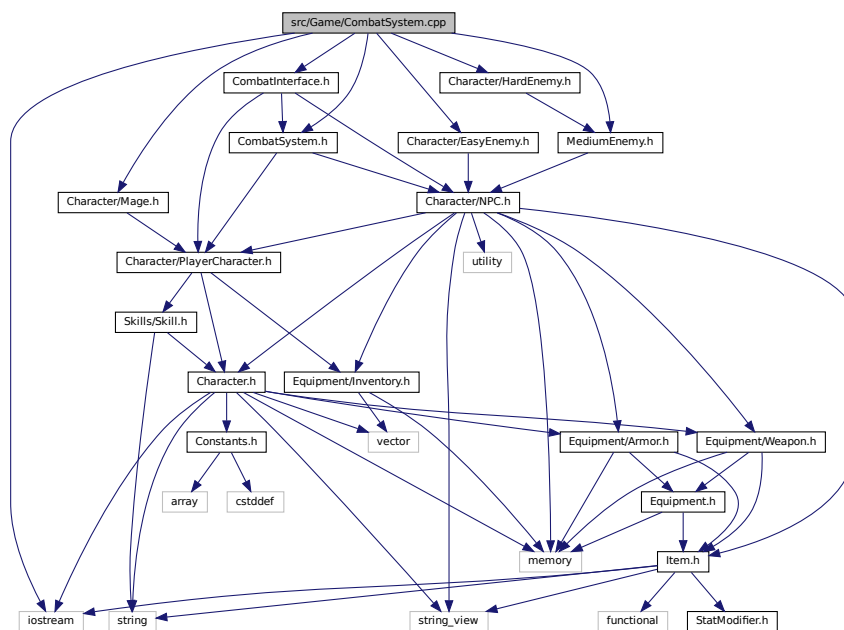
7.34 src/Game/CombatSystem.cpp File Reference

```

#include "CombatSystem.h"
#include "CombatInterface.h"
#include "Character/EasyEnemy.h"
#include "Character/HardEnemy.h"
#include "Character/MediumEnemy.h"
#include "Character/Mage.h"
#include <iostream>

```

Include dependency graph for `CombatSystem.cpp`:

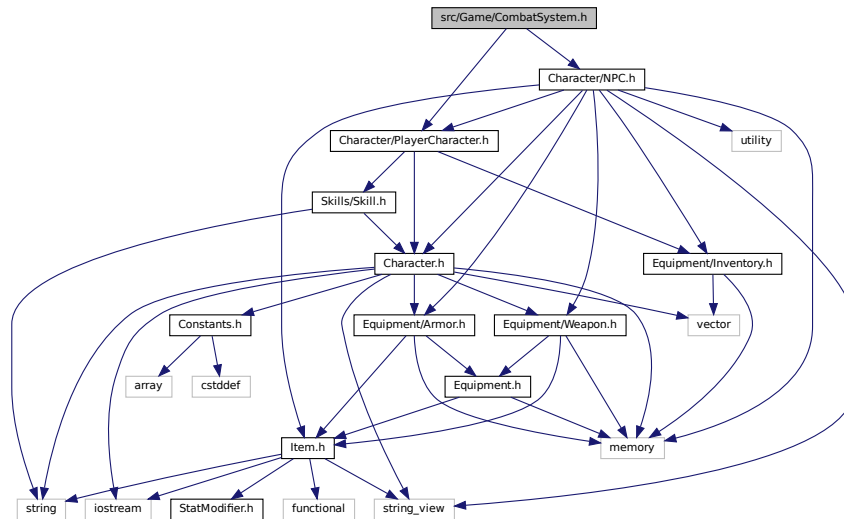


7.35 src/Game/CombatSystem.h File Reference

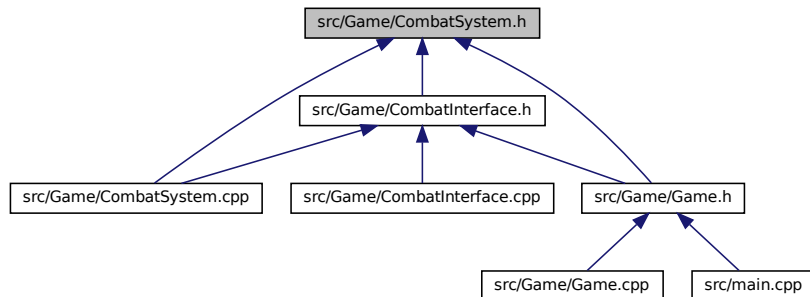
```
#include "Character/PlayerCharacter.h"
```

```
#include "Character/NPC.h"
```

Include dependency graph for CombatSystem.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [CombatSystem](#)

7.36 src/Game/computeStatBonus.cpp File Reference

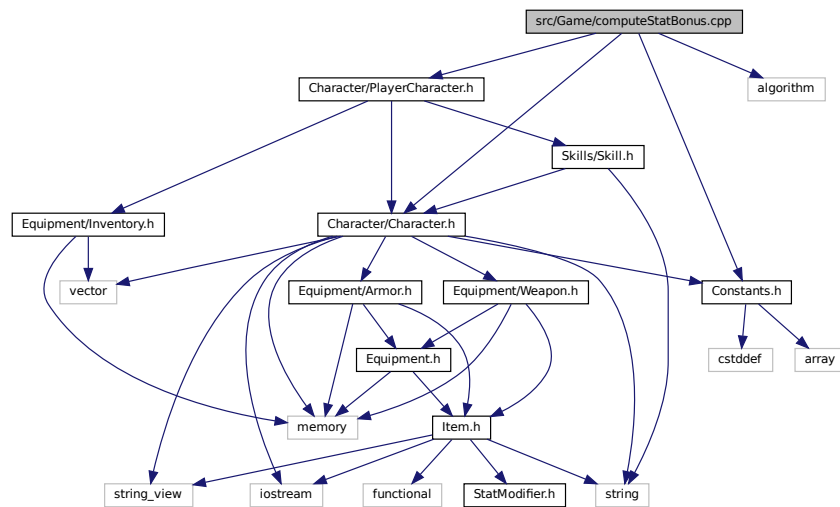
```
#include "Character/Character.h"
```

```
#include "Character/PlayerCharacter.h"
```

```
#include "Constants.h"
```

```
#include <algorithm>
```

Include dependency graph for computeStatBonus.cpp:



Functions

- int `computeStatBonus` (const `Character` &attacker, int damage, bool isMelee)

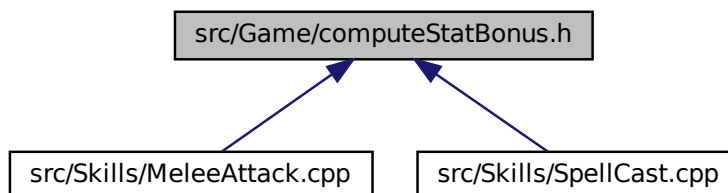
7.36.1 Function Documentation

7.36.1.1 `computeStatBonus()`

```
int computeStatBonus (
    const Character & attacker,
    int damage,
    bool isMelee )
```


7.37 src/Game/computeStatBonus.h File Reference

This graph shows which files directly or indirectly include this file:



Functions

- `int computeStatBonus (const Character &attacker, int damage, bool isMelee)`

7.37.1 Function Documentation

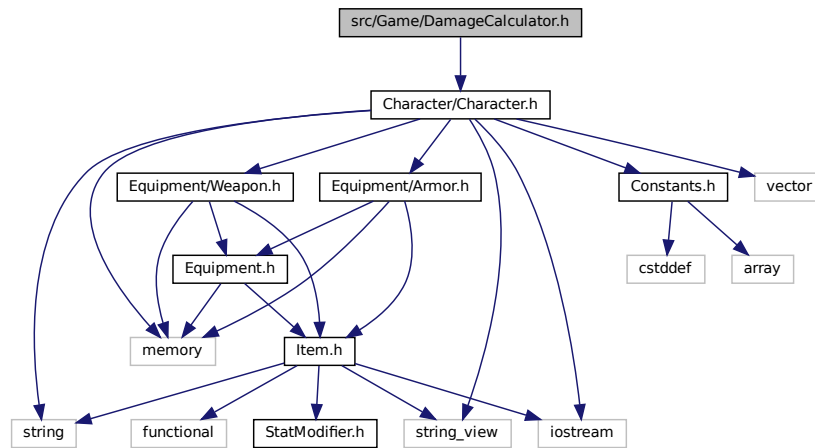
7.37.1.1 computeStatBonus()

```
int computeStatBonus (  
    const Character & attacker,  
    int damage,  
    bool isMelee )
```

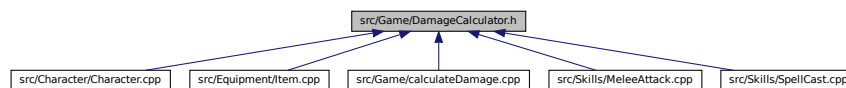
7.38 src/Game/DamageCalculator.h File Reference

```
#include "Character/Character.h"
```

Include dependency graph for DamageCalculator.h:



This graph shows which files directly or indirectly include this file:



Classes

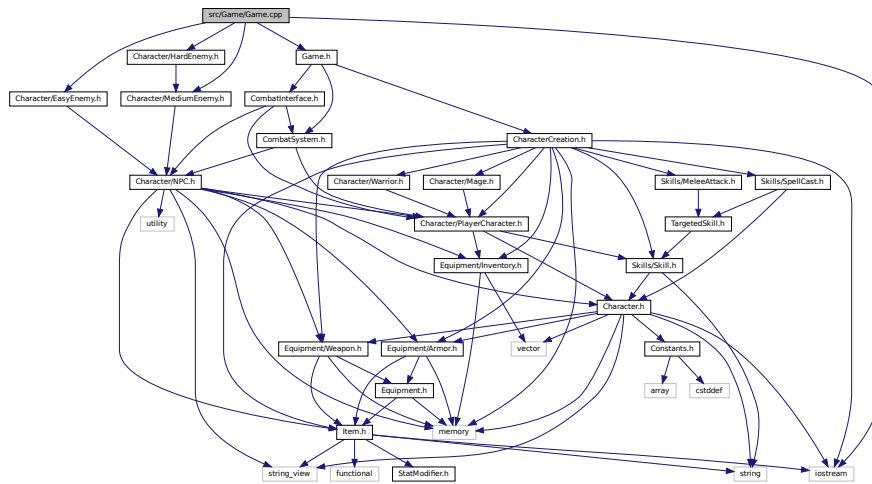
- class [DamageCalculator](#)

7.39 src/Game/Game.cpp File Reference

```
#include "Game.h"
#include "Character/EasyEnemy.h"
#include "Character/MediumEnemy.h"
#include "Character/HardEnemy.h"
```

```
#include <iostream>
```

Include dependency graph for Game.cpp:



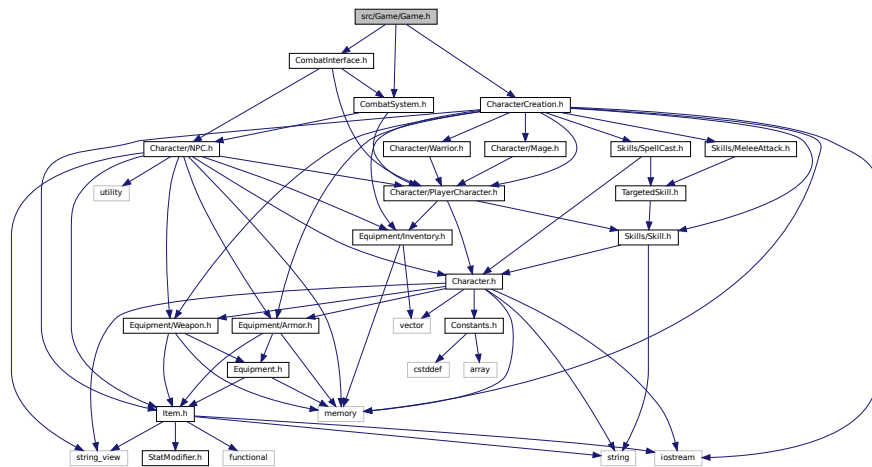
7.40 src/Game/Game.h File Reference

```
#include "CharacterCreation.h"
```

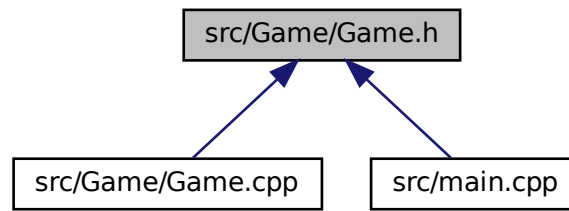
```
#include "CombatSystem.h"
```

```
#include "CombatInterface.h"
```

Include dependency graph for Game.h:



This graph shows which files directly or indirectly include this file:



Classes

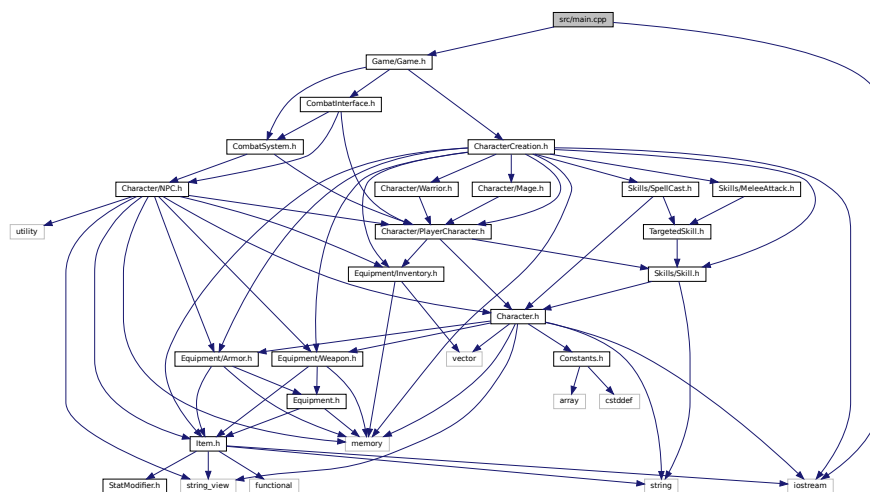
- class [Game](#)

7.41 src/main.cpp File Reference

```
#include "Game/Game.h"
```

```
#include <iostream>
```

Include dependency graph for main.cpp:



Functions

- int [main](#) ()

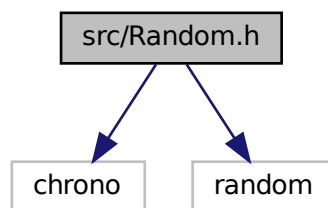
7.41.1 Function Documentation

7.41.1.1 main()

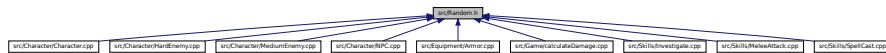
```
int main ( )
```

7.42 src/Random.h File Reference

```
#include <chrono>
#include <random>
Include dependency graph for Random.h:
```



This graph shows which files directly or indirectly include this file:



Namespaces

- [Random](#)

Functions

- `std::mt19937 Random::generate ()`
- `int Random::get (int min, int max)`
- `template<typename T >`
`T Random::get (T min, T max)`
- `template<typename R, typename S, typename T >`
`R Random::get (S min, T max)`

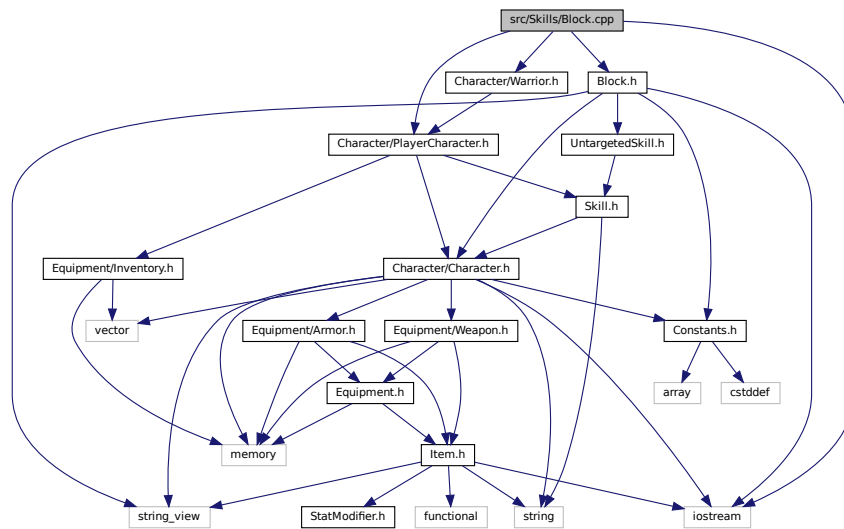
Variables

- `std::mt19937 Random::mt { generate() }`

7.43 src/Skills/Block.cpp File Reference

```
#include "Block.h"
#include "Character/PlayerCharacter.h"
#include "Character/Warrior.h"
#include <iostream>
```

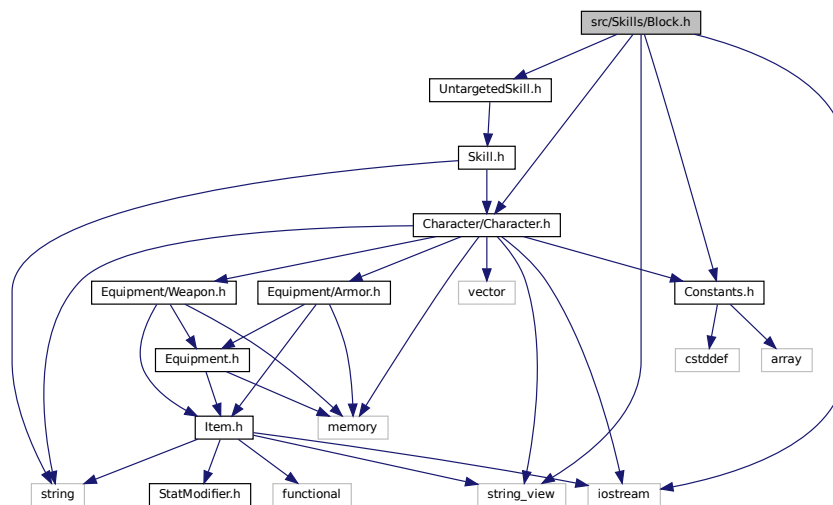
Include dependency graph for Block.cpp:



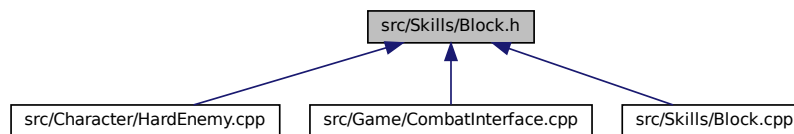
7.44 src/Skills/Block.h File Reference

```
#include "UntargetedSkill.h"
#include "Character/Character.h"
#include "Constants.h"
#include <string_view>
#include <iostream>
```

Include dependency graph for Block.h:



This graph shows which files directly or indirectly include this file:



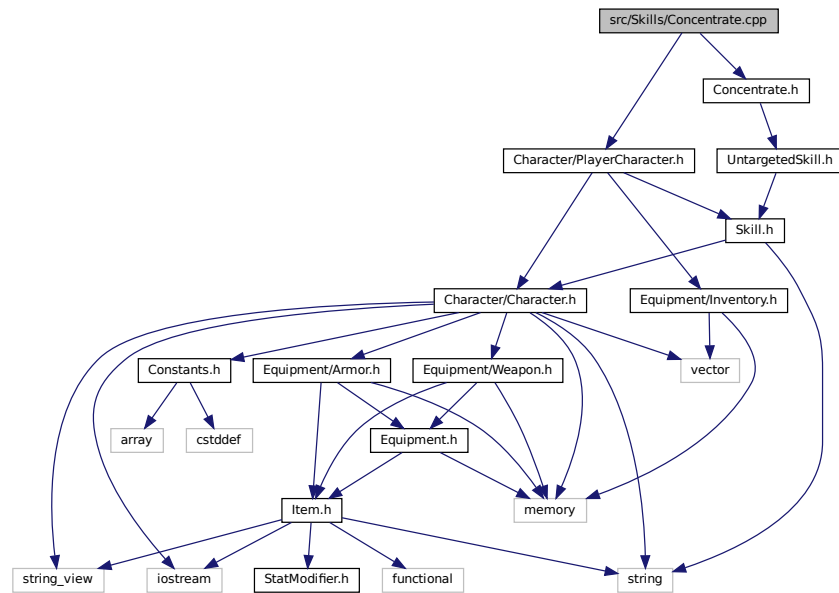
Classes

- class [Block](#)

7.45 src/Skills/Concentrate.cpp File Reference

```
#include "Concentrate.h"
#include "Character/PlayerCharacter.h"
```

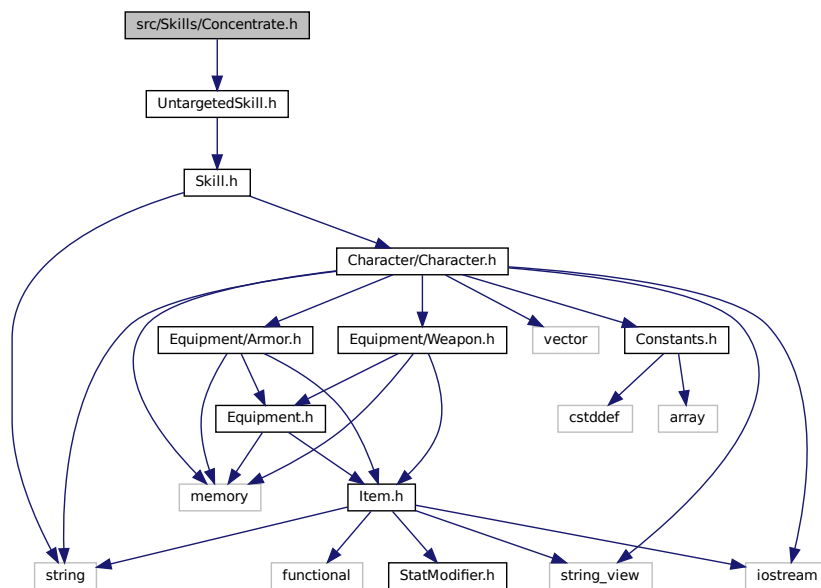
Include dependency graph for Concentrate.cpp:



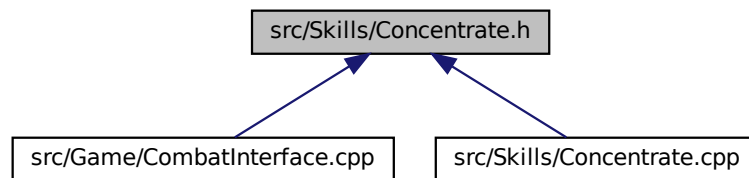
7.46 src/Skills/Concentrate.h File Reference

```
#include "UntargetedSkill.h"
```

Include dependency graph for Concentrate.h:



This graph shows which files directly or indirectly include this file:



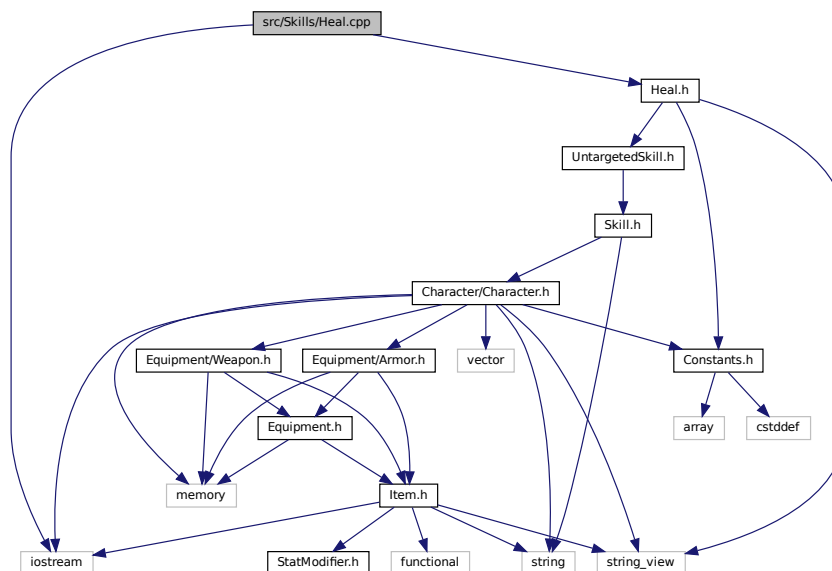
Classes

- class [Concentrate](#)

7.47 src/Skills/Heal.cpp File Reference

```
#include "Heal.h"
#include <iostream>
```

Include dependency graph for Heal.cpp:

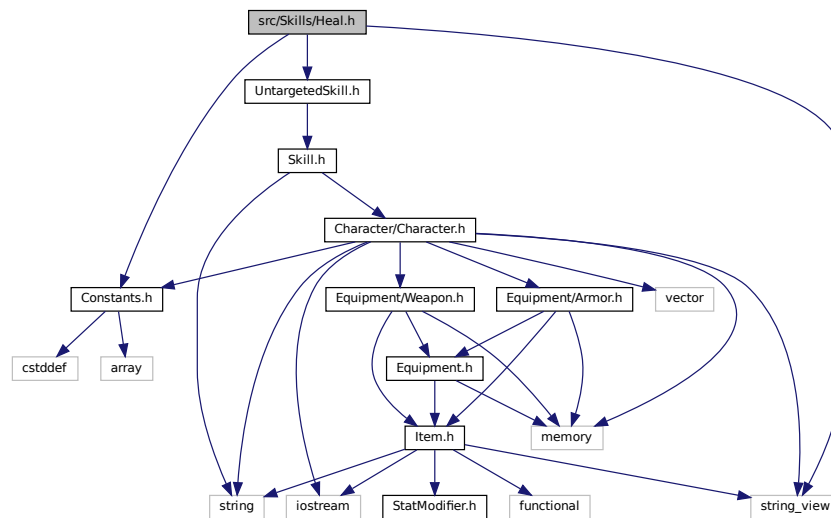


7.48 src/Skills/Heal.h File Reference

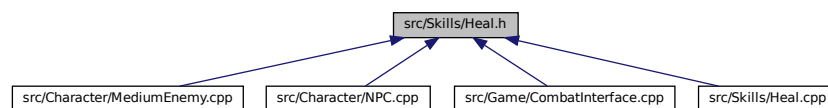
```
#include "Constants.h"
#include "UntargetedSkill.h"
```

```
#include <string_view>
```

Include dependency graph for Heal.h:



This graph shows which files directly or indirectly include this file:



Classes

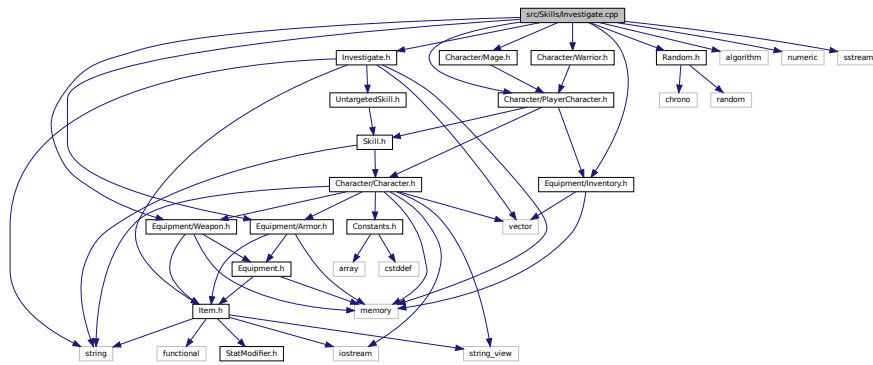
- class [Heal](#)

7.49 src/Skills/Investigate.cpp File Reference

```
#include "Investigate.h"
#include "Equipment/Armor.h"
#include "Equipment/Weapon.h"
#include "Equipment/Inventory.h"
#include "Character/PlayerCharacter.h"
#include "Character/Mage.h"
#include "Character/Warrior.h"
#include "Random.h"
#include <algorithm>
#include <numeric>
```

```
#include <sstream>
```

Include dependency graph for Investigate.cpp:



7.50 src/Skills/Investigate.h File Reference

```
#include "UntargetedSkill.h"
```

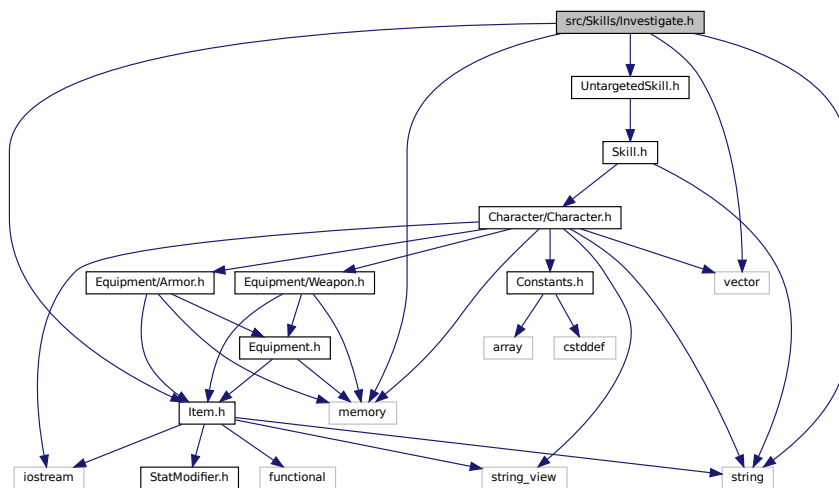
```
#include "Equipment/Item.h"
```

```
#include <memory>
```

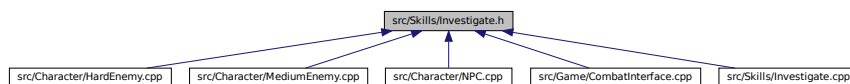
```
#include <vector>
```

```
#include <string>
```

Include dependency graph for Investigate.h:



This graph shows which files directly or indirectly include this file:



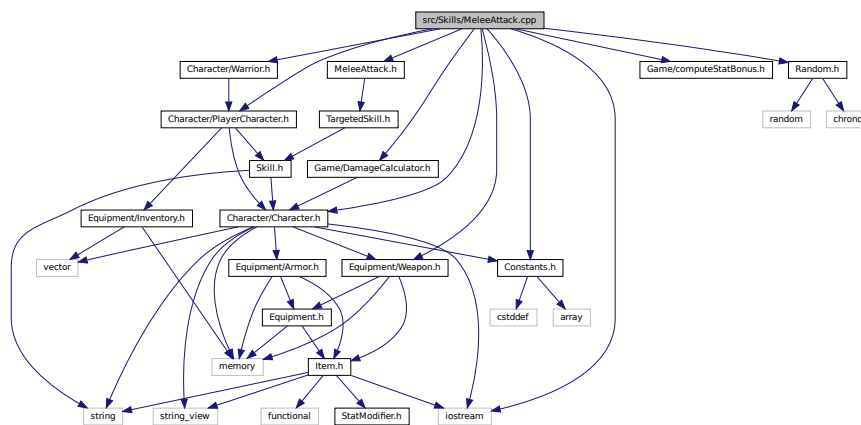
Classes

- class [Investigate](#)
- struct [Investigate::ModifierProfile](#)

7.51 src/Skills/MeleeAttack.cpp File Reference

```
#include "MeleeAttack.h"
#include "Character/Character.h"
#include "Character/PlayerCharacter.h"
#include "Character/Warrior.h"
#include "Equipment/Weapon.h"
#include "Game/computeStatBonus.h"
#include "Game/DamageCalculator.h"
#include "Random.h"
#include "Constants.h"
#include <iostream>
```

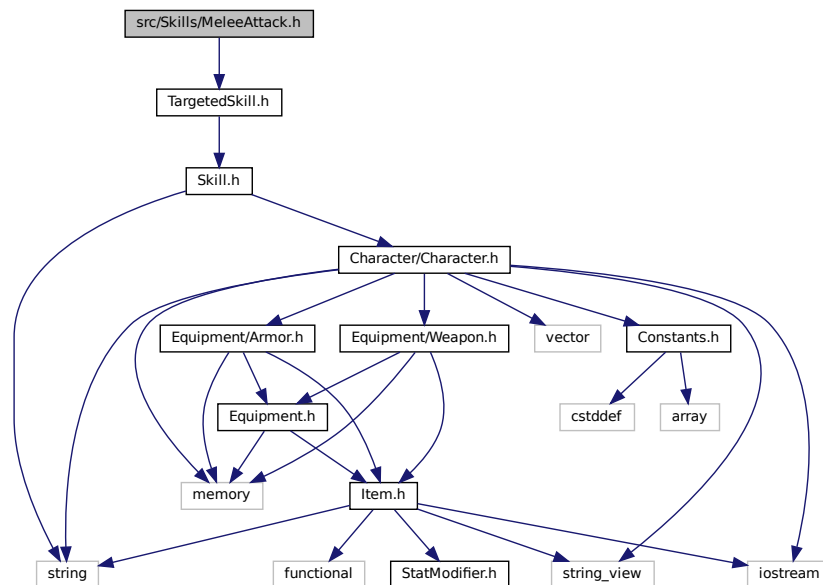
Include dependency graph for MeleeAttack.cpp:



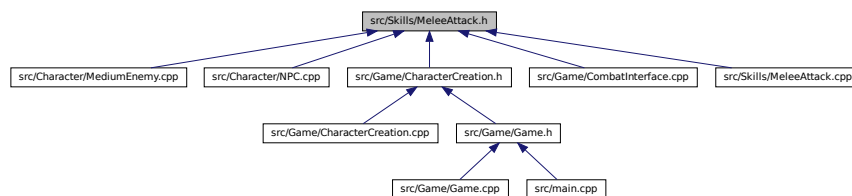
7.52 src/Skills/MeleeAttack.h File Reference

```
#include "TargetedSkill.h"
```

Include dependency graph for MeleeAttack.h:



This graph shows which files directly or indirectly include this file:



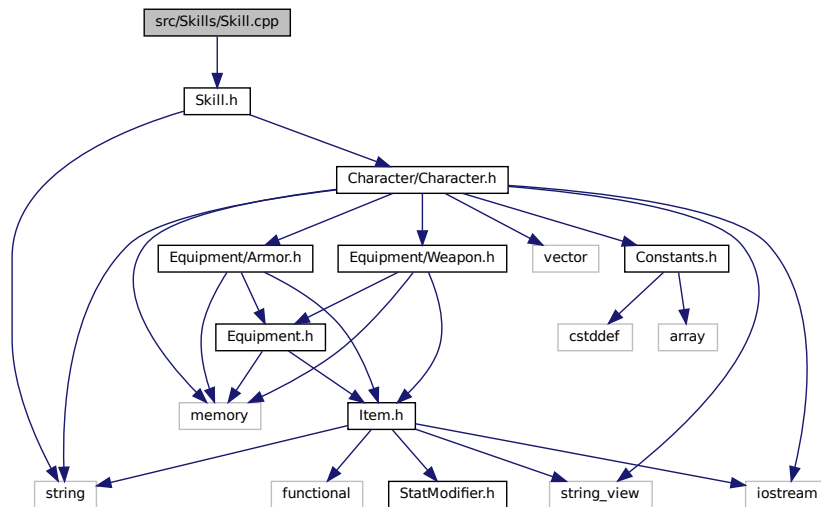
Classes

- class [MeleeAttack](#)

7.53 src/Skills/Skill.cpp File Reference

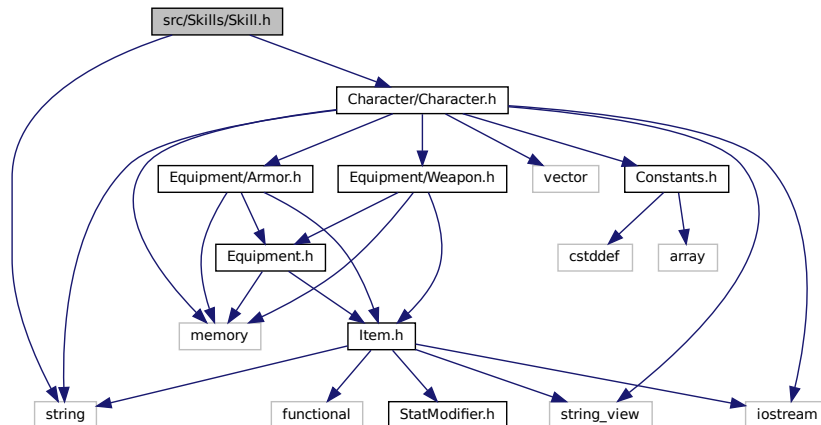
```
#include "Skill.h"
```

Include dependency graph for Skill.cpp:

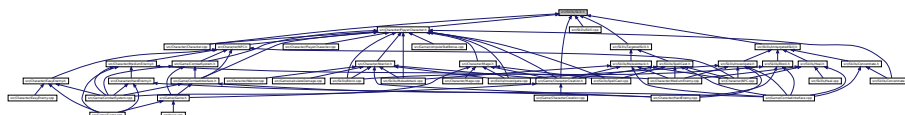


7.54 src/Skills/Skill.h File Reference

```
#include <string>
#include "Character/Character.h"
Include dependency graph for Skill.h:
```



This graph shows which files directly or indirectly include this file:



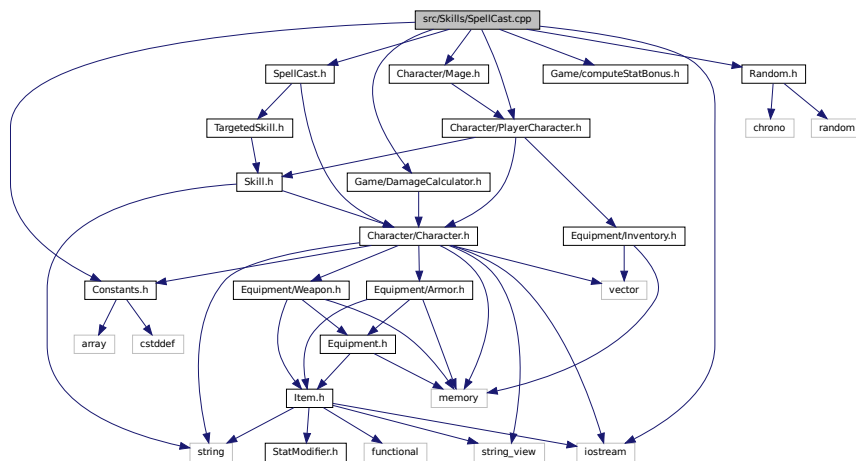
Classes

- class [Skill](#)

7.55 src/Skills/SpellCast.cpp File Reference

```
#include "SpellCast.h"
#include "Character/PlayerCharacter.h"
#include "Character/Mage.h"
#include "Game/computeStatBonus.h"
#include "Game/DamageCalculator.h"
#include "Random.h"
#include "Constants.h"
#include <iostream>
```

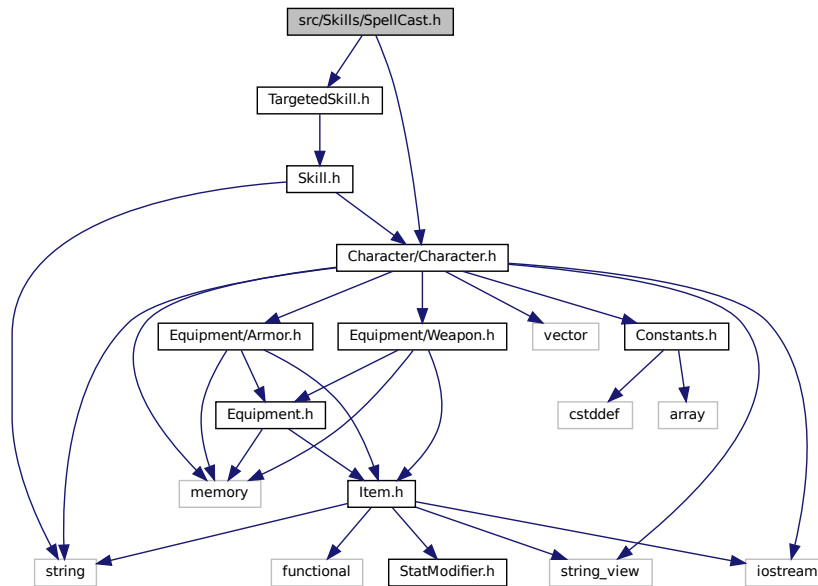
Include dependency graph for SpellCast.cpp:



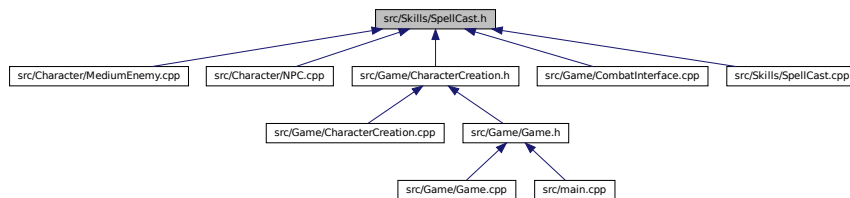
7.56 src/Skills/SpellCast.h File Reference

```
#include "TargetedSkill.h"
#include "Character/Character.h"
```

Include dependency graph for SpellCast.h:



This graph shows which files directly or indirectly include this file:



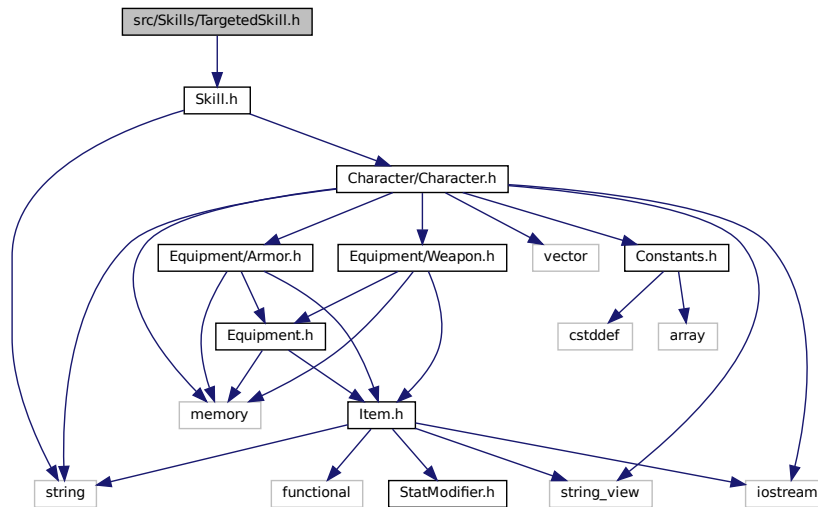
Classes

- class [SpellCast](#)

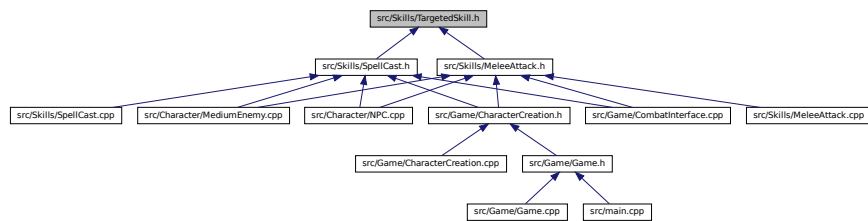
7.57 src/Skills/TargetedSkill.h File Reference

```
#include "Skill.h"
```


Include dependency graph for TargetedSkill.h:



This graph shows which files directly or indirectly include this file:



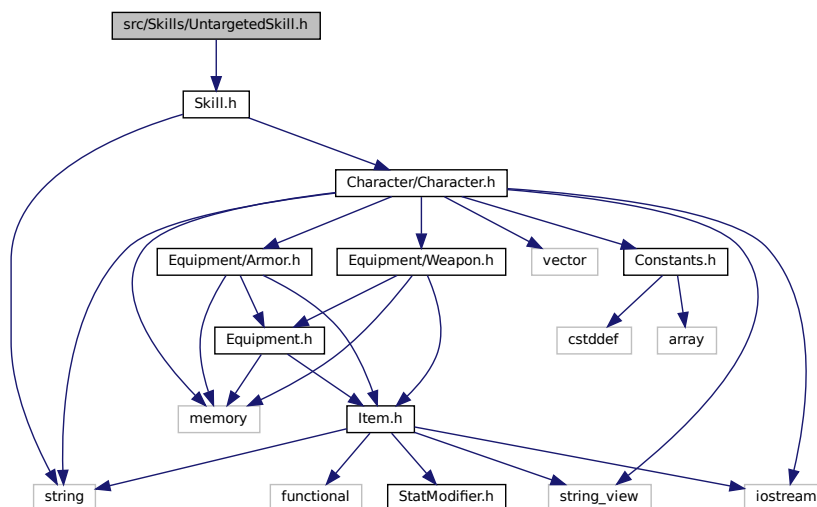
Classes

- class [TargetedSkill](#)

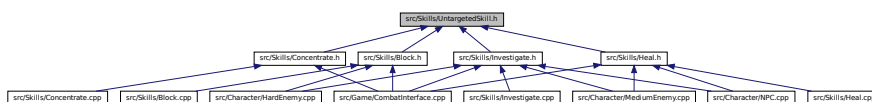
7.58 src/Skills/UntargetedSkill.h File Reference

```
#include "Skill.h"
```

Include dependency graph for UntargetedSkill.h:



This graph shows which files directly or indirectly include this file:



Classes

- class [UntargetedSkill](#)

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